

A.U.R.A.

Autonomous Universal Routing Architecture

Combined Technical Design Document — v2.0

DSP, Microphone, NAS Integration & Smart Room Extension

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Abstract

A.U.R.A. (Autonomous Universal Routing Architecture) is a modular, desk-mounted audio ecosystem designed to unify three independent audio sources — WiFi-streamed IoT audio, Bluetooth A2DP, and legacy optical S/PDIF from a Sony PlayStation 2 or television — into a single high-fidelity amplification chain feeding Sony MHC-GX2000 speakers. The system requires zero manual source selection: an ESP32-based signal-presence detection engine continuously monitors all three input paths and automatically routes the active source through a relay switching stage to the final amplifier.

This v2.0 document extends the v1.1 TDD with three major new subsystem specifications: a DSP & Audio Processing Subsystem (Section 10) covering two-tier IIR/FIR signal processing, room correction, speaker protection, soft limiting, and loudness compensation; a Microphone & Room Correction Subsystem (Section 11) covering calibrated acoustic measurement, MEMS I2S microphone integration, on-device wake-word detection, and voice control command architecture; and a NAS Integration Subsystem (Section 12) covering the full self-hosted software stack including CamillaDSP room correction, Navidrome and Jellyfin media servers, Lyrion Music Server for multi-room audio, and a fully local AI assistant pipeline (whisper.cpp + Ollama + Piper TTS + Home Assistant).

All v1.1 subsystems (WiFi Streaming, Bluetooth, Optical/Legacy, Analog Convergence, Power Distribution, Intelligent Source Routing) are carried forward unchanged. All v1.1 supplements (ADC threshold derivation, block diagrams, power budget, contention exit logic, series resistors, gap closure tracking) are retained. Sections 8 and 9 are updated with the expanded roadmap and extended reference list respectively.

System Architecture Overview

A.U.R.A. is built around a three-domain separation philosophy borrowed from professional audio engineering: the High-Voltage Power Domain, the Digital Transport Domain, and the Low-Level Analog Domain are kept physically and electrically isolated from each other to guarantee zero cross-domain interference. This is not merely wiring — it is a Multi-Domain Signal Chain.

A. Power Domain

- Source: 19V laptop adapter (minimum 4A — see Supplement C for derivation) → splitter.
- Path 1 (Muscle Rail): 19V direct to TPA3116D2 amplifier — preserves dynamic headroom for bass transients.
- Path 2 (Brain Rail): 19V → LM2596 buck converter → 5V regulated logic rail feeding ESP32, PCM5102A, Optical DAC, and Cubetek hub.
- Why: The amplifier benefits from the full 19V supply voltage. Sensitive digital logic requires a clean, regulated 5V rail free of switching noise from the muscle rail.

B. Digital Transport Domain

- IoT Path: WiFi → ESP32 → I2S protocol → PCM5102A DAC → analog out.
- Bluetooth Path: BT device → Cubetek (A2DP receive mode) → Optical (Toslink) → Optical DAC → analog out.
- Legacy Path: PS2 / TV → Optical cable → Cubetek (S/PDIF passthrough mode) → Optical DAC → analog out.
- Why: I2S is a bit-perfect, clock-synchronised protocol. Optical S/PDIF provides galvanic isolation — electrical noise from the PS2's motor and fan cannot physically cross the optical medium.

C. Analog Convergence Layer

- Merge point: PCM5102A RCA output (via 10 kΩ series resistor) + Optical DAC RCA output (via 10 kΩ series resistor) → DPDT toggle switch. See Supplement E for impedance protection detail.
- Final stage: DPDT switch → TPA3116D2 master volume knob → Sony MHC-GX2000 speakers.
- Auto-switching: ESP32 monitors signal presence on all input paths and drives the DPDT (or relay in v2) to the active source — no manual intervention required.

System Block Diagram

Layer	Primary Hardware	MCU Interface	Function
WiFi Streaming	ESP32, PCM5102A 32-bit DAC	I2S (BCLK, LRCLK, DOUT, MCLK)	WiFi audio decode, I2S push to DAC

Layer	Primary Hardware	MCU Interface	Function
Bluetooth	Cubetek hub (A2DP RX mode)	Cubetek internal — optical out	BT A2DP receive, output to Optical DAC
Optical/Legacy	Cubetek (S/PDIF pass), Optical DAC	Cubetek mode select — optical chain	PS2/TV galvanically isolated audio path
Analog Convergence	10 k Ω series R, DPDT switch, TPA3116D2 amp	ESP32 relay drive (v2) / manual (v1)	Impedance isolation, source select, Class-D amplification
Power	19V brick, LM2596 buck, star GND	VCC, GND, star ground node	Rail isolation, logic regulation, inrush
Source Routing	ESP32 signal-presence firmware	ADC signal monitors, relay GPIO	Auto source detection and switching

Section 1 — WiFi Streaming Subsystem

1.1 Overview

The WiFi Streaming Subsystem is the primary IoT audio input path of A.U.R.A. It uses the ESP32 microcontroller to receive audio data over a local WiFi network and deliver it to the PCM5102A Digital-to-Analog Converter via the I2S (Inter-IC Sound) protocol. The result is a bit-perfect, low-latency audio stream with 32-bit depth and up to 384 kHz sample rate — exceeding CD quality by a significant margin.

1.2 Design Philosophy

1.2.1 Why I2S

I2S is a dedicated serial bus standard for digital audio. Unlike UART or SPI, I2S carries a dedicated word-clock (LRCLK) that keeps left/right channels perfectly phase-aligned and a bit-clock (BCLK) that ensures every sample is clocked in at the correct moment. This makes it bit-perfect — no resampling, no jitter artefacts. The ESP32 has a hardware I2S peripheral with DMA support, meaning audio data is transferred to the PCM5102A without CPU intervention, freeing the ESP32 to run signal-presence monitoring and source routing concurrently.

1.2.2 Why PCM5102A over the internal ESP32 DAC

The ESP32's built-in DAC is a simple 8-bit converter with high distortion and a noise floor unsuitable for speaker-level audio. The PCM5102A is a specialist converter with 112 dB SNR, 32-bit resolution, and a dedicated internal oscillator that removes the need for an external MCLK in most configurations. The difference is audible: the internal DAC hisses; the PCM5102A is silent at idle.

1.2.3 MCLK Mode Decision

The PCM5102A can operate in two clock modes: Master mode (it generates MCLK internally from BCLK via PLL) and Slave mode (MCU provides MCLK). For A.U.R.A. v1.0, the PCM5102A is configured in Master clock mode with the SCK pin tied to 3.3V, enabling its internal PLL. This eliminates the need for a fourth I2S wire and reduces jitter from the ESP32's software clock generation.

1.3 Hardware Architecture

1.3.1 Signal Chain

Stage	Component	Function
WiFi receive	ESP32 (802.11 b/g/n)	Receives audio stream via UDP/TCP on local network
Audio decode	ESP32 firmware	Decodes MP3/AAC/PCM depending on stream format
I2S output	ESP32 I2S peripheral (DMA)	Clocks digital audio out on BCLK/LRCLK/DOUT
DAC conversion	PCM5102A (32-bit / 384 kHz)	Converts I2S digital stream to 2.1V RMS analog

Stage	Component	Function
Impedance isolation	10 kΩ series resistor (NEW v1.1)	Prevents mutual DAC loading at DPDT switch — see Supplement E
Analog output	RCA line out (L/R)	Feeds DPDT source selector switch

1.3.2 I2S Pin Mapping — ESP32 to PCM5102A

Signal	ESP32 GPIO	PCM5102A Pin	Description
BCLK (Bit Clock)	GPIO 26	BCK (Pin 13)	Clocks each individual bit of audio data
LRCLK (Word Clock)	GPIO 25	LRCK (Pin 15)	Selects Left or Right channel per sample
DOUT (Data)	GPIO 22	DIN (Pin 14)	Serial audio data stream (MSB first)
SCK (System Clock)	Tied to 3.3V	SCK (Pin 12)	3.3V = internal PLL master clock mode
FMT	Tied to GND	FMT (Pin 16)	GND = I2S standard format (not left-justified)
DEMP	Tied to GND	DEMP (Pin 8)	GND = de-emphasis filter disabled
XSMT (Soft Mute)	GPIO 27 (optional)	XSMT (Pin 17)	Logic HIGH = normal play, LOW = soft mute
VCC	3.3V rail	VCC (Pin 1)	Logic supply — 3.3V from brain rail
GND	Star GND node	GND (Pin 2)	Must connect to system star ground point

1.3.3 PCM5102A Key Specifications

Parameter	Value
Resolution	32-bit
Maximum sample rate	384 kHz
SNR (Signal-to-Noise Ratio)	112 dB (A-weighted)
THD+N (Total Harmonic Distortion)	-93 dB typical
Output level	2.1V RMS (standard line level)
Logic supply	3.3V (ESP32 compatible — no level shifting required)
Clock mode (v1.0)	Internal PLL — SCK tied to 3.3V
Output coupling	Capacitor-coupled RCA line out (L/R)
Output impedance	~100 Ω (TI datasheet) — protected by 10 kΩ series R at DPDT switch (Supplement E)

1.4 Software Implementation

1.4.1 Core Firmware Loop

- Step 1: On boot, ESP32 connects to the configured WiFi SSID and starts a UDP or TCP audio server on a fixed port.
- Step 2: A sending device (phone, laptop, Raspberry Pi) streams PCM audio data over the local network to the ESP32's IP address.
- Step 3: The ESP32 DMA engine feeds the incoming audio buffer directly into the I2S peripheral without CPU copying.
- Step 4: The I2S peripheral clocks out BCLK, LRCLK, and DOUT to the PCM5102A at the configured sample rate.
- Step 5: Concurrently, the signal-presence monitor checks for audio activity on the I2S stream and asserts the WiFi-active flag to the source routing engine.

1.4.2 Key Firmware Parameters

Parameter	Value	Rationale
I2S sample rate	44,100 Hz (default)	CD-quality baseline; upgradeable to 96 kHz
I2S bit depth	32-bit	PCM5102A native — no conversion overhead
DMA buffer count	8 buffers	Prevents underrun on WiFi packet jitter
DMA buffer length	512 samples	Balances latency (~11.6 ms) vs dropout risk
WiFi mode	Station (STA)	Connects to existing home router
Audio protocol	UDP (low latency)	No retransmit delay; dropouts acceptable for audio
Signal presence threshold	> -60 dBFS RMS	Distinguishes active stream from silent buffer — see Supplement A for ADC derivation

1.5 Bill of Materials — WiFi Streaming Subsystem

Qty	Component	Value / Part	Purpose	Supplier
1	Microcontroller	ESP32-WROOM-32 / DevKit	WiFi receive, I2S master, source routing MCU	Robu.in
1	DAC Module	PCM5102A I2S DAC board	32-bit I2S to analog conversion	AliExpress / Robu.in
1	Capacitor	100 nF ceramic	PCM5102A VCC decoupling	Amazon
2	Series resistors (NEW v1.1)	10 kΩ (L + R channels)	Output impedance protection at DPDT switch — Supplement E	Amazon

Qty	Component	Value / Part	Purpose	Supplier
2	RCA connectors	Female RCA (L/R)	Analog line-level output to DPDT switch	Local
1	Cable	3.5mm stereo to RCA (shielded OFC)	PCM5102A out to DPDT switch	Amazon
4	Jumper wire	22 AWG solid	I2S pin connections (BCLK, LRCLK, DOUT, GND)	Local

1.6 Engineering Assessment — WiFi Streaming Subsystem

Strengths

- PCM5102A 112 dB SNR is best-in-class for this price tier — orders of magnitude cleaner than the ESP32 internal DAC.
- I2S DMA transfers audio without CPU intervention, freeing the ESP32 for source routing logic simultaneously.
- Internal PLL clock mode eliminates a fourth wire and reduces jitter compared to software-generated MCLK.
- 3.3V logic compatibility means no level shifting between ESP32 and PCM5102A.
- Modular: the ESP32 can be swapped for a Raspberry Pi Zero 2W with zero changes to the PCM5102A circuit.
- 10 kΩ series output resistors (v1.1 addition) prevent mutual loading between idle and active DAC paths at the DPDT switch.

Limitations

- UDP streaming has no retransmit — a bad WiFi packet causes a brief audio dropout. Acceptable for music; unacceptable for latency-critical applications.
- WiFi interference can cause buffer underruns on congested 2.4 GHz networks. Mitigation: use 5 GHz band if ESP32 variant supports it, or increase DMA buffer count.
- The LM2596 buck converter switching noise (~150 kHz) can appear on the PCM5102A power rail. Mitigation: add a 10 μH ferrite bead + 100 nF ceramic on the VCC trace to the PCM5102A.

1.7 Academic Terminology Reference — WiFi Streaming Subsystem

Technique	Formal Description
I2S Protocol	Inter-IC Sound: a synchronous serial bus with dedicated word-clock and bit-clock lines ensuring phase-aligned stereo audio transfer with no resampling.
DMA Transfer	Direct Memory Access: hardware-level buffer transfer from RAM to I2S peripheral without CPU instruction cycles, enabling concurrent task execution.

Technique	Formal Description
PCM5102A PLL	Phase-Locked Loop master clock: internal oscillator derives system clock from BCLK, eliminating external MCLK wire and reducing clock jitter.
SNR (112 dB)	Signal-to-Noise Ratio: logarithmic ratio of signal power to noise floor. 112 dB corresponds to a noise floor approximately 400,000× below full-scale signal amplitude.
Signal Presence Detection	RMS energy thresholding on the I2S audio buffer: determines whether an active audio stream is present to inform the source routing engine.
Output Impedance Loading (NEW)	When two DAC outputs share a passive merge node, each idle source's output impedance forms an AC load on the active source, reducing output voltage and increasing distortion. Series resistors eliminate this mutual loading path.

Section 2 — Bluetooth Subsystem

2.1 Overview

The Bluetooth Subsystem allows any Bluetooth A2DP-capable device — phones, tablets, laptops — to stream audio wirelessly to A.U.R.A. without requiring the device to be on the same WiFi network. This path uses the Cubetek hub as an A2DP Bluetooth receiver, which then outputs the received audio over Toslink optical to the Optical DAC, maintaining galvanic isolation all the way to the analog convergence stage.

2.2 Design Philosophy

2.2.1 Why Cubetek for Bluetooth (v1.0)

The ESP32 supports Bluetooth Classic A2DP natively, but in v1.0 the ESP32's processing budget is reserved for WiFi streaming, I2S output, and source routing logic. Running simultaneous WiFi and Bluetooth on the ESP32 is possible but introduces RF co-existence interference on the 2.4 GHz band, increasing audio dropout probability on both paths. The Cubetek hub offloads Bluetooth reception to dedicated hardware, eliminating this interference entirely.

2.2.2 Why Optical Output from Cubetek

The Cubetek outputs received Bluetooth audio over Toslink optical to the Optical DAC. This choice is deliberate: the Bluetooth audio — even though it originated as a wireless signal — is converted to S/PDIF optical before entering the analog domain. This means any RF interference or power supply noise from the Bluetooth radio circuit is galvanically isolated before reaching the sensitive analog output stage. The Optical DAC then performs a clean digital-to-analog conversion at the same RCA line level as the PCM5102A.

2.2.3 Cubetek Dual-Mode Operation

The Cubetek hub is capable of receiving both Bluetooth A2DP audio and PS2/TV optical S/PDIF simultaneously. It outputs only the currently selected mode through its Toslink output. In A.U.R.A. v1.0, mode selection is handled by the Cubetek's own mode switch. In v2.0, the ESP32's source routing engine will assert a GPIO line to the Cubetek's mode select pin, enabling fully automatic mode arbitration without physical intervention.

2.3 Hardware Architecture

2.3.1 Signal Chain

Stage	Component	Function
BT transmit	Any A2DP device (phone/laptop)	Streams audio over Bluetooth Classic A2DP
BT receive	Cubetek hub (A2DP RX mode)	Receives and decodes A2DP audio stream
Digital output	Cubetek → Toslink optical cable	S/PDIF optical output (galvanically isolated)

Stage	Component	Function
DAC conversion	Optical DAC (Toslink to RCA)	192 kHz / 24-bit S/PDIF to RCA analog
Impedance isolation	10 kΩ series resistor (NEW v1.1)	Prevents mutual DAC loading at DPDT switch
Analog output	RCA line out (L/R)	Feeds DPDT source selector switch

2.3.2 Cubetek Key Operating Parameters

Parameter	Value
Bluetooth version	Bluetooth 5.0 (A2DP profile)
Supported codecs	SBC, AAC (device dependent)
Output format	Toslink optical S/PDIF
Power supply	5V DC (from LM2596 brain rail)
Mode selection	Hardware mode switch (v1.0) / ESP32 GPIO (v2.0 roadmap)
Dual-mode capability	Receives BT and PS2 optical simultaneously; outputs selected mode only

2.3.3 Optical DAC Key Specifications

Parameter	Value
Input	Toslink optical (S/PDIF)
Maximum sampling rate	192 kHz / 24-bit
Output	RCA analog (L/R), standard line level
Power supply	5V DC via DC barrel or Micro-USB (from brain rail)
Output impedance	~600 Ω typical — protected by 10 kΩ series R at DPDT (Supplement E)
Purpose in this path	Converts Cubetek Bluetooth optical output to analog for DPDT switch

2.4 Inter-Subsystem Communication

The Bluetooth subsystem interfaces with the rest of A.U.R.A. entirely through the analog domain: its RCA output feeds the DPDT switch alongside the PCM5102A output from the WiFi path. The ESP32 source routing engine detects Bluetooth activity by monitoring the analog signal level on the Optical DAC RCA output (via a simple voltage divider to an ESP32 ADC pin). When Bluetooth audio is detected and WiFi is silent, the ESP32 drives the DPDT switch to the Optical DAC input. No manual source selection is required.

2.5 Bill of Materials — Bluetooth Subsystem

Qty	Component	Value / Part	Purpose	Supplier
1	Bluetooth Hub	Cubetek BT/Optical hub	A2DP receive + optical output	Amazon / local
1	Optical DAC	Toslink to RCA DAC (192kHz/24-bit)	S/PDIF optical to RCA analog	Amazon / AliExpress
1	Toslink cable	1m optical (dust-capped)	Cubetek output to Optical DAC	Amazon
1	Cable	RCA to RCA (shielded OFC)	Optical DAC output to DPDT switch	Amazon
2	Series resistors (NEW v1.1)	10 k Ω (L + R channels)	Output impedance isolation at DPDT — Supplement E	Amazon
2	Resistors (voltage divider)	10 k Ω + 1 k Ω	Voltage divider for ESP32 signal-presence ADC tap	Amazon

2.6 Engineering Assessment — Bluetooth Subsystem

Strengths

- Dedicated Cubetek hardware eliminates 2.4 GHz co-existence interference between WiFi and Bluetooth.
- Optical output from Cubetek maintains galvanic isolation through the digital transport stage.
- Cubetek dual-mode operation means a single device handles both BT and PS2 optical.
- Clean upgrade path: v2.0 migrates BT to ESP32 A2DP stack, freeing Cubetek exclusively for PS2 optical.

Limitations

- Bluetooth A2DP SBC codec introduces lossy compression. Mitigated if the source device supports AAC.
- Cubetek mode selection is manual in v1.0. Automatic ESP32 GPIO mode control is a v2.0 feature.
- Bluetooth pairing must be done once per new device — no automatic pairing with multiple remembered devices simultaneously.

2.7 Academic Terminology Reference — Bluetooth Subsystem

Technique	Formal Description
A2DP Profile	Advanced Audio Distribution Profile: Bluetooth profile defining the unidirectional streaming of stereo audio from a source device to a sink (receiver).

Technique	Formal Description
SBC Codec	Sub-Band Coding: mandatory lossy codec for A2DP; compresses audio by dividing the frequency spectrum into sub-bands and quantising each independently.
S/PDIF Optical	Sony/Philips Digital Interface over optical fibre (Toslink): transmits PCM audio as light pulses, providing galvanic isolation by removing electrical conductors from the signal path.
Galvanic Isolation	Complete electrical separation between two circuit domains via a non-conductive medium (optical fibre). Prevents ground loops and noise injection across the isolation barrier.
RF Co-existence	Interference between concurrent wireless transceivers on overlapping frequency bands (WiFi 802.11 and Bluetooth both operate at 2.4 GHz). Avoided in v1.0 by hardware separation.

Section 3 — Optical / Legacy Subsystem

3.1 Overview

The Optical/Legacy Subsystem connects the Sony PlayStation 2 (and any S/PDIF-capable television or legacy device) to A.U.R.A. using an all-optical signal chain. The defining engineering goal of this path is galvanic isolation: the PS2's internal power supply, motor controllers, and cooling fan generate significant electrical noise that, if allowed to travel through a copper cable, would introduce audible hum and hiss into the speaker output. Optical S/PDIF (Toslink) eliminates this problem entirely — noise cannot cross a beam of light.

3.2 Design Philosophy

3.2.1 Why Not a 3.5mm or RCA Cable from the PS2

The PS2's analog audio output is sourced from its internal DAC, which shares a power rail with the disc motor, optical pickup, and GPU. These high-current, noisy components couple interference onto the power rail, which then appears as noise on the analog output. A copper cable carries this noise directly to the amplifier. The PS2's optical output, by contrast, is electrically inert — it is a modulated light beam that carries only digital audio data.

3.2.2 The Cubetek as S/PDIF Hub

Rather than connecting the PS2 optical output directly to the Optical DAC, the signal is routed through the Cubetek hub first. This enables the Cubetek to act as a single Toslink output point for both the PS2/legacy path and the Bluetooth path, feeding the same Optical DAC regardless of which source is active. This reduces the Optical DAC count from two to one and simplifies the wiring topology.

3.3 Hardware Architecture

3.3.1 Signal Chain

Stage	Component	Function
Source	Sony PS2 / TV optical output	Generates S/PDIF digital audio over Toslink
Input cable	Toslink optical cable #1 (1m)	Carries PS2 optical output to Cubetek
Hub / mode select	Cubetek hub (S/PDIF passthrough mode)	Routes PS2 optical to Toslink output
Output cable	Toslink optical cable #2 (1m)	Carries Cubetek output to Optical DAC
DAC conversion	Optical DAC (192 kHz / 24-bit)	S/PDIF to RCA analog conversion
Impedance isolation	10 kΩ series resistor (NEW v1.1)	Shared with BT path; prevents DPDT loading
Analog output	RCA line out (L/R)	Feeds DPDT source selector switch

3.3.2 Optical DAC Key Specifications

Parameter	Value
Input connector	Toslink (IEC 60958 / S/PDIF)
Maximum input rate	192 kHz / 24-bit
Output connector	RCA (L/R)
Output level	~2.0V RMS (standard line level, matching PCM5102A)
Power supply	5V DC from LM2596 brain rail
Supported formats	PCM stereo (2-channel). Dolby Digital / DTS pass-through (not decoded)
PS2 output format	PCM stereo at 44.1 kHz / 16-bit (standard for PS2 game audio)

3.4 Galvanic Isolation — Engineering Detail

Galvanic isolation via optical fibre ensures that the PS2's electrical noise domain and A.U.R.A.'s analog domain share no conductive path. Specifically:

- The PS2's motor controller and optical pickup draw pulsed currents of several amperes from its internal PSU. These transients cause voltage spikes on the PS2's internal ground plane.
- In a copper-connected system, these ground spikes would travel to the amplifier's GND via the cable shield, appearing as hum or broadband noise on the speakers.
- With Toslink, the PS2's electrical domain ends at its LED transmitter. A.U.R.A.'s electrical domain begins at the Optical DAC's photodiode receiver. The only link between them is a photon — which carries no electrical noise.
- Measured result: zero audible noise contribution from the PS2 path at idle when using Toslink, verified by listening test and oscilloscope on the Optical DAC RCA output.

3.5 Bill of Materials — Optical/Legacy Subsystem

Qty	Component	Value / Part	Purpose	Supplier
1	Cubetek hub	Cubetek BT/Optical hub	S/PDIF passthrough, shared with BT subsystem	Amazon / local
2	Toslink cable	1m optical (dust-capped)	PS2→Cubetek and Cubetek→Optical DAC	Amazon
1	Optical DAC	Toslink to RCA (192kHz/24-bit)	S/PDIF to analog, shared with BT path	Amazon / AliExpress
1	Cable	RCA to RCA (shielded OFC)	Optical DAC RCA out to DPDT switch	Amazon

Note: The Cubetek hub and Optical DAC appear in both Section 2 (Bluetooth) and Section 3 (Optical/Legacy) BOMs as they are physically shared. They appear once in the Consolidated BOM in Appendix A. The 10 k Ω series resistors for this path are shared with Section 2 BOM.

3.6 Engineering Assessment — Optical/Legacy Subsystem

Strengths

- Complete galvanic isolation eliminates all PS2-sourced electrical noise.
- The Cubetek hub consolidates Bluetooth and PS2 optical paths onto a single Toslink output.
- 192 kHz / 24-bit Optical DAC exceeds PS2's 44.1 kHz / 16-bit output — future-proofed.
- Zero software dependency: the optical path functions correctly even if the ESP32 firmware is not running.

Limitations

- PS2 audio is limited to 44.1 kHz / 16-bit PCM — the Optical DAC's 192 kHz capability is underutilised on this source.
- Cubetek manual mode selection in v1.0 — automatic switching requires v2.0 firmware.
- Two Toslink cables introduce two additional optical connector interfaces, each a potential point of signal loss if dust-capped cables are not used.

Section 4 — Analog Convergence Layer

4.1 Overview

The Analog Convergence Layer is where all three digital paths — WiFi (PCM5102A), Bluetooth (Optical DAC via Cubetek), and Optical/Legacy (Optical DAC via Cubetek) — meet at the same physical point and one is selected for amplification. This layer consists of three components: a pair of 10 kΩ series output protection resistors (one per DAC path, added in v1.1), a DPDT toggle switch acting as the source selector, and the TPA3116D2 Class-D amplifier driving the Sony MHC-GX2000 speakers.

4.2 Output Impedance Protection — Series Resistors (NEW v1.1)

In v1.0, both DAC outputs were connected directly to the DPDT switch with no isolation between them. When one source was idle, its output stage presented a low-impedance AC path to ground, loading the active source and slightly reducing its output voltage and increasing distortion. Two 10 kΩ series resistors — one on each RCA feed to the DPDT switch — break this mutual loading path. The DPDT switch then selects between two resistor-isolated feeds. See Supplement E for full derivation.

4.3 The DPDT Source Selector

4.3.1 Why DPDT

A standard single-pole switch disconnects only one wire, leaving the other channel's ground path intact. This creates a condition called crosstalk — the idle source bleeds faintly into the active source's output. A DPDT (Double Pole Double Throw) switch simultaneously disconnects both the Left and Right signal lines when switching, ensuring complete isolation between the PCM5102A path and the Optical DAC path.

4.3.2 Input Configuration

DPDT Position	Source Connected	Signal Path
Position A	PCM5102A (WiFi Streaming)	ESP32 → I2S → PCM5102A → 10 kΩ series R → RCA → DPDT Pole A
Position B	Optical DAC (BT + PS2 shared)	Cubetek → Toslink → Optical DAC → 10 kΩ series R → RCA → DPDT Pole B

Note: Bluetooth and PS2/Optical share Position B via the Cubetek hub. The Cubetek performs the first layer of arbitration between BT and PS2 based on its mode selection. The DPDT switch then selects between the WiFi path and the Cubetek-aggregated multimedia path.

4.3.3 Auto-Switching in v1.0

In v1.0, the DPDT switch is a physical toggle switch. The ESP32 source routing engine (Section 6) detects which source is active and asserts an LED indicator to guide the user. In v2.0, the physical DPDT is replaced by an ESP32-driven relay module that performs source switching automatically without any physical interaction.

4.4 The TPA3116D2 Amplifier

4.4.1 Why Class-D

The TPA3116D2 is a Class-D switching amplifier with over 90% power efficiency. Unlike Class-AB amplifiers, which dissipate the difference between supply voltage and output voltage as heat, Class-D amplifiers use pulse-width modulation to switch the output transistors fully on or fully off — producing almost no heat at any output level. At idle with no music playing, the TPA3116D2 draws negligible current and generates zero heat. This makes it ideal for an always-on system.

4.4.2 Why 19V Supply

The TPA3116D2 is rated for 12V–26V operation. Running it at 19V (direct from the laptop adapter) provides dynamic headroom — the amplifier can deliver transient power peaks without clipping before the supply voltage limit is reached. This is audible as improved bass impact and cleaner transient response on percussive audio. The Sony MHC-GX2000 speakers are rated at 6Ω impedance and can handle the power output at 19V without risk of damage.

4.4.3 TPA3116D2 Key Specifications

Parameter	Value
IC	Texas Instruments TPA3116D2 (original TI chip — verify authenticity on purchase)
Supply voltage	DC 12V – 26V (19V laptop adapter direct)
Output power (6Ω load)	50W + 50W (standard) / up to 100W + 100W (BTL mode)
Efficiency	> 90% at typical listening levels
Idle current draw	< 50 mA (no signal)
Heat sink	Required — board must have integrated aluminium heat sink
Volume control	Pre-installed potentiometer on board (master volume knob)
Speaker terminals	Screw terminals (L/R+ and L/R–)
Input	RCA line level (2V RMS nominal — matches PCM5102A and Optical DAC output)
Input impedance	~10 kΩ typical — see Supplement E for voltage divider effect with series protection R

4.4.4 Sony MHC-GX2000 Speaker Interface

Parameter	Value
Speaker model	Sony MHC-GX2000 passive speakers
Impedance	6Ω
Connection	Screw terminals on TPA3116D2 board
Cable	18 AWG OFC speaker wire (minimum — lower gauge for longer runs)

Parameter	Value
Polarity	Match +/- markings on both TPA3116D2 screw terminals and speaker terminals

4.5 Star Grounding in the Analog Stage

The star grounding topology (described in full in Section 5) has a critical role in the analog convergence layer. All RCA cable shields, the DPDT switch chassis (if metal), and the TPA3116D2 GND terminal must converge at the same single physical ground point as the ESP32, PCM5102A, and Optical DAC. Any ground path that forms a loop — even a small one — will act as an antenna for mains-frequency interference (50 Hz in India), producing an audible hum on the speakers. Star grounding eliminates all ground loops by ensuring current has exactly one return path.

4.6 Bill of Materials — Analog Convergence Layer

Qty	Component	Value / Part	Purpose	Supplier
1	Amplifier board	TPA3116D2 (original TI chip, with heatsink and volume pot)	Class-D speaker amplification at 19V	Amazon / Robu.in
1	Source switch	DPDT toggle switch (rated > 1A)	Simultaneous L+R source selection	Amazon / electronics shop
4	Series resistors (NEW v1.1)	10 kΩ (L+R × 2 DAC paths)	Output impedance isolation at DPDT switch — Supplement E	Amazon
1	Power jack	DC barrel jack female 5.5mm × 2.1mm	19V laptop adapter input	Amazon
1	Splitter	5.5mm × 2.1mm DC power splitter (1× to 2×)	Splits 19V to amp and buck converter	Amazon
2m	Speaker wire	18 AWG OFC (2-conductor)	TPA3116D2 to Sony MHC-GX2000 speakers	Local
2	RCA cable	Shielded OFC RCA-to-RCA (0.5m)	DPDT output to TPA3116D2 input	Amazon

4.7 Engineering Assessment — Analog Convergence Layer

Strengths

- DPDT switch eliminates crosstalk by cutting both L and R channels simultaneously.
- 10 kΩ series resistors (v1.1 addition) prevent mutual DAC loading at the passive merge node, eliminating distortion from idle-source impedance.
- TPA3116D2 Class-D topology ensures zero idle heat and negligible idle current, making always-on operation practical.
- 19V supply gives meaningful dynamic headroom over a 12V supply — audible on bass-heavy content.

- Both input sources (PCM5102A and Optical DAC) output the same 2V RMS line level — no level mismatch or volume jump on switching.

Limitations

- Physical DPDT switch in v1.0 requires manual operation. Auto-switching relay in v2.0 resolves this.
- 10 k Ω series resistors form a voltage divider with the TPA3116D2's ~10 k Ω input impedance, attenuating the signal by approximately 6 dB. This is compensated by advancing the TPA3116D2 volume control by approximately one-third of travel. Full derivation in Supplement E.
- TPA3116D2 board quality varies significantly by vendor — insist on the original TI chip marking and avoid 'mini' boards without heat sinks.

Section 5 — Power Distribution Network

5.1 Overview

The Power Distribution Network (PDN) is the backbone of A.U.R.A. Its role is to ensure that the high-current transients generated by the TPA3116D2 amplifier during bass transients do not corrupt the logic supply to the ESP32 and DACs, and that the LM2596 buck converter's switching noise does not appear on the analog audio outputs. This is achieved through strict rail separation, star grounding, and a hierarchical decoupling strategy.

5.2 Split-Rail Architecture

5.2.1 Muscle Rail (19V — High Power)

- Source: 19V laptop DC adapter. Minimum 4A rating required — see Supplement C for worst-case current derivation.
- Load: TPA3116D2 Class-D amplifier only.
- Why direct: The amplifier benefits from the full supply voltage for dynamic headroom. No regulation is needed — the TPA3116D2 is designed for unregulated DC input.

5.2.2 Brain Rail (5V — Logic and Sensing)

- Converter: LM2596 adjustable buck converter module, set to exactly 5.00V using a multimeter.
- Load: ESP32 (240 mA peak WiFi), PCM5102A (15 mA), Optical DAC (200 mA), Cubetek hub (300 mA).
- Total brain rail load: approximately 755 mA — well within the LM2596's 3A rated output.
- Multiturn potentiometer: mandatory for precise 5.0V calibration. Single-turn pots cannot be dialled accurately enough — 5.3V on the ESP32 VCC is a failure mode.

5.3 Current Budget

Component	Typical Current	Peak Current	Rail
ESP32 (WiFi active)	80 mA	240 mA	5V Brain
PCM5102A DAC	15 mA	20 mA	5V Brain (3.3V via onboard reg)
Optical DAC	150 mA	200 mA	5V Brain
Cubetek hub	200 mA	300 mA	5V Brain
Brain rail total	~445 mA	~760 mA	5V (LM2596 3A max)
TPA3116D2 (idle)	50 mA	50 mA	19V Muscle
TPA3116D2 (active, 6Ω)	~1,500 mA avg	~3,000 mA transient	19V Muscle
19V adapter peak load (NEW v1.1)	See Supplement C	~3,240 mA	19V — minimum 4A adapter required

The 19V adapter worst-case current calculation is derived in full in Supplement C. The v1.0 document specified a minimum 3A adapter, which is insufficient under simultaneous peak brain rail and amplifier transient draw. v1.1 corrects this to a minimum 4A adapter (65W minimum).

5.4 Star Grounding Topology

Star grounding is the single most important noise-prevention technique in A.U.R.A. All power return currents from every component — TPA3116D2, LM2596, ESP32, PCM5102A, Optical DAC, Cubetek — converge at one single physical point: the DC barrel jack's negative terminal. From there, a single heavy conductor returns to the adapter.

Why this matters: the TPA3116D2 draws up to 3A peak from the 19V rail during bass transients. If the amplifier's ground return shares a conductor with the PCM5102A's ground, the transient current spike creates a voltage drop across that shared conductor ($V = I \times R$). This appears as a noise voltage on the PCM5102A's GND pin — which is the reference for its analog output — and manifests as audible distortion or hiss. Star grounding prevents this by giving each component its own direct ground return path to the star point.

Ground Wire	AWG	From	To (Star Point)
TPA3116D2 return	18 AWG	TPA3116D2 GND terminal	DC barrel jack GND
LM2596 input GND	20 AWG	LM2596 Vin GND	DC barrel jack GND
LM2596 output GND	22 AWG	LM2596 Vout GND	DC barrel jack GND (same node)
ESP32 GND	24 AWG	ESP32 GND pin	LM2596 Vout GND node
PCM5102A GND	24 AWG	PCM5102A GND pin	LM2596 Vout GND node
Optical DAC GND	24 AWG	Optical DAC GND	LM2596 Vout GND node
Cubetek GND	24 AWG	Cubetek GND	LM2596 Vout GND node

5.5 Decoupling Hierarchy

Tier	Component	Placement	Function	Frequency Range
1 — Bulk	1000 μ F electrolytic (low-ESR)	Across 19V rail, near TPA3116D2	Absorbs amplifier transient current demand	DC to ~1 kHz
2 — Rail filter	100 μ F electrolytic + 100 nF ceramic	LM2596 output (5V rail)	Filters buck converter switching ripple	1 kHz to ~500 kHz
3 — Local MCU	100 nF ceramic	Directly across ESP32 VCC/GND pins	Suppresses fast switching noise at consumption point	500 kHz to several MHz

Tier	Component	Placement	Function	Frequency Range
4 — DAC supply	10 μ H ferrite bead + 100 nF ceramic	PCM5102A VCC trace	Isolates LM2596 switching ripple from DAC analog stage	150 kHz (LM2596 switching freq)

5.6 LM2596 Switching Noise — Engineering Note

The LM2596 operates at approximately 150 kHz switching frequency. This generates a ripple voltage on the 5V output rail that, if unchecked, can appear as a high-frequency whine on the PCM5102A audio output — audible as a faint whistle in quiet passages. The LM2596 module includes an output filter inductor and capacitor, but the PCM5102A's supply should be further isolated with a 10 μ H ferrite bead and 100 nF bypass capacitor on its VCC pin. This forms a low-pass filter that attenuates the 150 kHz component before it reaches the DAC.

5.7 Bill of Materials — Power Distribution Network

Qty	Component	Value / Part	Purpose	Supplier
1	Buck converter	LM2596 adjustable module (multiturn pot)	19V → 5V brain rail regulation	Amazon
1	Power jack	DC barrel female 5.5mm × 2.1mm	19V adapter input	Amazon
1	Power splitter	5.5mm × 2.1mm 1-to-2 DC splitter	Split 19V to amp and buck	Amazon
1	19V adapter (NEW v1.1)	Minimum 4A / 65W (revised from 3A)	Source adapter — 3A is insufficient; see Supplement C	Local / online
1	Electrolytic cap	1000 μ F 25V low-ESR	Bulk reservoir on 19V muscle rail	Robu.in
2	Electrolytic cap	100 μ F 16V	LM2596 output filter	Robu.in
10+	Ceramic cap	100 nF 50V	MCU and DAC local decoupling	Amazon
2	Ferrite bead	10 μ H (optional upgrade)	DAC VCC switching noise filter	Robu.in
1m	Power wire	18 AWG silicone (red/black)	Muscle rail and TPA3116D2 connections	Local
1m	Signal wire	24 AWG (black)	Star ground network	Local

5.8 Engineering Assessment — PDN

Strengths

- Split-rail architecture with direct 19V to amplifier and regulated 5V brain rail directly addresses the primary failure mode of logic corruption during amplifier transients.
- Star grounding prevents ground loop hum and transient-induced noise on the analog output.
- Hierarchical four-tier decoupling covers the full frequency range from DC supply variation to high-frequency MCU switching.
- LM2596 3A capacity provides 4× headroom over the measured 755 mA brain rail load.

Limitations

- v1.0 minimum adapter specification of 3A is insufficient at worst-case combined load. Corrected to 4A minimum in v1.1. See Supplement C.
- LM2596 efficiency is approximately 75–80% at 19V input / 5V output. A modern converter (TPS62xxx series) would achieve ~90% efficiency.
- LM2596 switching noise at 150 kHz requires additional filtering on the PCM5102A VCC trace. Without the ferrite bead upgrade, a faint whine may be audible at very high volume with no music playing.
- Star grounding requires careful physical wire routing — a single common-impedance ground path defeats the topology.

Section 6 — Intelligent Source Routing (Auto-Switching)

6.1 Overview

The Intelligent Source Routing (ISR) engine is the feature that transforms A.U.R.A. from a manually switched audio selector into a fully autonomous system. The user never touches a switch: A.U.R.A. detects which source is producing audio and routes it to the amplifier automatically. The routing decision is based on signal-presence detection — whichever source has an active audio signal is selected.

6.2 Design Philosophy

6.2.1 Signal-Presence Detection — Not Priority

A priority-based system (e.g., WiFi always wins over Bluetooth) would interrupt ongoing playback inappropriately. A.U.R.A. uses signal-presence detection instead: the currently playing source continues to play until it goes silent, at which point the system re-evaluates and switches to whichever other source has become active.

6.2.2 Hysteresis — Preventing Rapid Toggling

Audio sources regularly produce brief silences (gaps between tracks, menu transitions in games). Without hysteresis, a 200 ms silence would trigger a source switch, cutting back incorrectly. A.U.R.A. requires a source to be continuously silent for a defined hold-off period (default: 3 seconds) before switching. Similarly, a new source must produce audio continuously for a confirmation window (default: 500 ms) before being selected.

6.3 Hardware Architecture

6.3.1 Signal-Presence Sensing Circuit

Each audio source's analog RCA output is monitored by a simple voltage divider feeding an ESP32 ADC pin. The divider reduces the 2V RMS line level to within the ESP32's 0–3.3V ADC input range. See Supplement A for full threshold derivation.

Source	Sensing Point	Divider	ESP32 ADC Pin	Threshold
WiFi (PCM5102A)	PCM5102A RCA output (pre-DPDT)	10 kΩ + 3.3 kΩ	GPIO 34 (ADC1_CH6)	> 50 ADC counts RMS (≈ 40 mV — see Supplement A)
Multimedia (Optical DAC)	Optical DAC RCA output (pre-DPDT)	10 kΩ + 3.3 kΩ	GPIO 35 (ADC1_CH7)	> 50 ADC counts RMS (≈ 40 mV — see Supplement A)

Note: GPIO 34 and 35 are input-only ADC pins on the ESP32 — ideal for this application as they cannot accidentally drive the audio signal line.

6.3.2 DPDT / Relay Control (v1.0 vs v2.0)

Version	Switch Type	Control Method	User Interaction
v1.0 (current)	Physical DPDT toggle switch	Manual — ESP32 drives indicator LED only	User toggles switch; LED shows active source
v2.0 (roadmap)	SPDT relay module (5V coil)	ESP32 GPIO drives relay coil via transistor	Fully automatic — no user interaction required

6.4 Software Implementation

6.4.1 Source Routing State Machine

State	Condition	ESP32 Action	LED Indicator
IDLE	No source active (both silent for > 3s)	Relay centred (last source)	Both LEDs off
WIFI_ACTIVE	PCM5102A signal present, Optical DAC silent	Assert DPDT/relay to WiFi position	Blue LED on
MULTIMEDIA_ACTIVE	Optical DAC signal present, PCM5102A silent	Assert DPDT/relay to Optical DAC position	Green LED on
CONTENTION	Both sources active simultaneously	Hold current source; start 3s re-evaluation timer	Both LEDs blink
CONTENTION_RESOLVE (NEW v1.1)	Contention timer expires; both still active	Apply precedence rule: WiFi > Multimedia; switch to WiFi; log event	Blue LED on (solid)

The *CONTENTION_RESOLVE* state is a new addition in v1.1. In v1.0 the *CONTENTION* state had no defined exit condition for sustained simultaneous activity. The full contention exit logic is defined in Supplement D.

6.4.2 Detection Algorithm

- Step 1: Sample ADC on GPIO 34 (WiFi) and GPIO 35 (Multimedia) at 100 Hz (every 10 ms).
- Step 2: Compute a 500 ms rolling RMS window on each channel (50 samples).
- Step 3: If either RMS exceeds the threshold (50 ADC counts $\approx$ 40 mV), that source is classified as ACTIVE. See Supplement A for threshold derivation.
- Step 4: Apply hysteresis — a source must be ACTIVE for 500 ms continuously to trigger a switch TO it; it must be SILENT for 3 seconds continuously to trigger a switch AWAY from it.
- Step 5: On confirmed source change, assert the relay GPIO (v2.0) or illuminate the indicator LED (v1.0).
- Step 6: In CONTENTION state, hold the current source and start a 3-second re-evaluation timer. If both sources remain active when the timer expires, apply CONTENTION_RESOLVE precedence rule (WiFi > Multimedia). See Supplement D.

6.4.3 Key Firmware Parameters

Parameter	Value	Rationale
ADC sample rate	100 Hz (every 10 ms)	Fast enough to detect audio onset within one cycle of a 50 Hz signal
RMS window	500 ms (50 samples)	Smooths short silences between notes/words
Signal threshold	> 50 ADC counts RMS	≈ 40 mV at ADC pin; see Supplement A for full derivation
Activation hold-off	500 ms continuous	Prevents false trigger from brief noise burst
Deactivation hold-off	3 seconds continuous silence	Tolerates gaps between tracks and menu transitions
Contention hold	3 seconds, then CONTENTION_RESOLVE	Avoids cutting audio; resolves sustained simultaneous activity — Supplement D
Contention precedence (NEW)	WiFi > Multimedia	WiFi is primary IoT path; PS2/BT is secondary — consistent with use-case priority
Relay driver	2N2222 NPN transistor + 1N4007 flyback diode (v2.0)	Safe inductive load switching for relay coil

6.5 Full End-to-End Flow

- A phone connects to A.U.R.A. over WiFi and begins streaming audio. The ESP32 receives the stream, pushes it via I2S to the PCM5102A, and simultaneously detects audio presence on GPIO 34.
- After 500 ms of confirmed audio, the ESP32 asserts the WiFi source state. In v1.0, a blue LED lights. In v2.0, the relay switches to the PCM5102A input. Audio plays through the TPA3116D2 to the Sony speakers.
- The phone stops playing. After 3 seconds of silence, the ESP32 moves to IDLE.
- The user starts a PS2 game. The PS2 outputs optical S/PDIF → Cubetek → Optical DAC → RCA. The ESP32 detects this on GPIO 35.
- After 500 ms of confirmed audio, the ESP32 switches to MULTIMEDIA_ACTIVE.
- The user pauses the game for 2.9 seconds (under deactivation hold-off). The system stays in MULTIMEDIA_ACTIVE. When the game resumes, no re-evaluation occurs.
- The user pauses the game for longer than 3 seconds. The system transitions to IDLE.
- WiFi and PS2 both become active simultaneously. The system enters CONTENTION, holds the current source, and starts the 3-second timer. If both remain active after 3 seconds, CONTENTION_RESOLVE switches to WiFi. See Supplement D.

6.6 Bill of Materials — Source Routing

Qty	Component	Value / Part	Purpose	Supplier
2	Resistor	10 k Ω	Signal-presence voltage divider (upper leg, $\times 2$ channels)	Amazon
2	Resistor	3.3 k Ω	Signal-presence voltage divider (lower leg, $\times 2$ channels)	Amazon
2	LED	5mm (blue + green)	Source active indicators (WiFi / Multimedia)	Amazon
2	Resistor	220 Ω	LED current limiting	Amazon
1	Relay module	5V SPDT relay (v2.0 roadmap)	Automatic source switching	Amazon / Robu.in
1	Transistor	2N2222 NPN (v2.0 roadmap)	Relay coil driver	Amazon
1	Diode	1N4007 (v2.0 roadmap)	Relay coil flyback protection	Amazon

Section 7 — Firmware and Validation

7.1 Overview

This section documents the hardware validation sketches and firmware modules developed for A.U.R.A. v1.0. These are written for the Arduino ESP32 core and the Arduino IDE. Test sketches are hardware validation tools — they are not production code. Their purpose is to verify that individual hardware stages function correctly before integration into the combined firmware.

7.2 Test Sketch 1 — PCM5102A I2S Tone Generator

Check	Expected
Serial output	'I2S tone test running...' on 115200 baud
Oscilloscope on PCM5102A RCA out	Clean 1 kHz sine wave, ~2V peak-to-peak, no noise floor visible
Audio monitor	Clear 1 kHz tone, zero hiss or distortion
Pass criterion	RCA output present and clean on both L and R channels simultaneously

7.3 Test Sketch 2 — WiFi Audio Stream Receive

Check	Expected
Serial output	ESP32 IP address printed; 'Listening for UDP audio on port 4444'
Audio test	Stream 44.1 kHz PCM WAV file from PC using audio streaming tool
Speaker output	Clean audio, no dropout for > 60 seconds on uncongested 2.4 GHz network
ADC monitor (GPIO 34)	ADC counts > 50 when audio playing, < 10 when silent — see Supplement A for threshold derivation
Pass criterion	Audio plays without dropout; signal-presence ADC responds correctly

7.4 Test Sketch 3 — Signal Presence ADC Verification

Check	Expected
Serial output	CSV format: 'WiFi_ADC, Multimedia_ADC' at 100 Hz
WiFi source active (PCM5102A playing)	GPIO 34 reads 50–200 ADC counts; GPIO 35 reads 0–10

Check	Expected
Multimedia source active (Optical DAC playing)	GPIO 35 reads 50–200 ADC counts; GPIO 34 reads 0–10
Both silent	Both channels read < 10 ADC counts
Pass criterion	Clean separation between active and silent state on each channel

7.5 Pin Mapping Reference Table

Function	ESP32 GPIO	Connected To	Section Reference
I2S BCLK	GPIO 26	PCM5102A BCK (Pin 13)	Sec 1.3
I2S LRCLK	GPIO 25	PCM5102A LRCK (Pin 15)	Sec 1.3
I2S DOUT	GPIO 22	PCM5102A DIN (Pin 14)	Sec 1.3
PCM5102A XSMT (mute)	GPIO 27	PCM5102A XSMT (Pin 17)	Sec 1.3
WiFi signal-presence ADC	GPIO 34 (input only)	10 kΩ + 3.3 kΩ divider from PCM5102A RCA	Sec 6.3, Supp A
Multimedia signal-presence ADC	GPIO 35 (input only)	10 kΩ + 3.3 kΩ divider from Optical DAC RCA	Sec 6.3, Supp A
WiFi source LED	GPIO 2	220 Ω + blue LED to GND	Sec 6.4
Multimedia source LED	GPIO 4	220 Ω + green LED to GND	Sec 6.4
Relay control (v2.0)	GPIO 5	2N2222 base via 1 kΩ; relay coil	Sec 6.3
3.3V supply	3.3V pin	PCM5102A VCC (via onboard regulator)	Sec 1.3
5V supply	VIN pin	LM2596 5V brain rail	Sec 5.2
GND	GND pin	Star ground node (LM2596 Vout GND)	Sec 5.4

7.6 Validation Status Summary

Subsystem	Status	Notes
PCM5102A I2S wiring	PASS	1 kHz tone test confirmed clean analog output on both channels
ESP32 WiFi connectivity	PASS	Connects to home router; IP address printed on serial
WiFi audio streaming	PASS	44.1 kHz PCM audio plays without dropout on local network

Subsystem	Status	Notes
PCM5102A MCLK internal PLL	PASS	SCK tied to 3.3V; no external MCLK required; clean output
Optical DAC (Toslink to RCA)	PASS	PS2 optical output converts to clean RCA analog
Cubetek BT receive mode	PASS	Pairs with phone; audio plays through Optical DAC chain
Cubetek S/PDIF passthrough mode	PASS	PS2 Toslink routes through Cubetek to Optical DAC correctly
DPDT switch operation	PASS	Switches both L+R simultaneously; no crosstalk between positions
10 kΩ series resistors (NEW v1.1)	PASS	Bench verified: no mutual loading between idle and active DAC paths
TPA3116D2 amplifier	PASS	19V supply; clean audio on Sony MHC-GX2000 speakers at all volumes
Star ground (hiss test)	PASS	No audible hiss at maximum volume with no audio source
Signal presence ADC (GPIO 34/35)	PASS	Correct ADC readings for active and silent states on both channels
Source routing state machine	IN PROGRESS	State transitions verified in bench test; relay integration pending
CONTENTION_RESOLVE logic (NEW v1.1)	IN PROGRESS	WiFi precedence logic implemented; sustained contention bench test pending
Auto-switching relay (v2.0)	PENDING	Relay module not yet wired; DPDT manual switch in use for v1.0
LM2596 output voltage calibration	PASS	Set to 5.00V using multimeter; confirmed with DMM
19V adapter capacity (NEW v1.1)	PENDING	Minimum 4A adapter procured per Supplement C; load test pending

Section 8 — Version Roadmap

8.1 v1.0 / v1.1 — Implemented (Current Hardware)

Full specification of the implemented v1.1 system is documented in Sections 1–7 and Supplements A–F. Key implemented features:

- ESP32 WiFi streaming → PCM5102A I2S DAC → TPA3116D2 → Sony MHC-GX2000 speakers.
- Cubetek A2DP Bluetooth receive → Toslink → Optical DAC → TPA3116D2.
- Sony PS2 optical (Toslink) → Cubetek passthrough → Optical DAC → TPA3116D2.
- DPDT manual source selector with ESP32 signal-presence detection (GPIO 34/35, 10 kΩ + 3.3 kΩ divider).
- Star-grounded 19V/5V split-rail power distribution (minimum 4A adapter — Supplement C).
- 10 kΩ series output impedance protection resistors on both DAC paths (Supplement E).
- CONTENTION_RESOLVE state machine with WiFi precedence (Supplement D).

8.2 v2.0 — Active Development

Feature	Section	Priority	Status	Notes
Auto-switching relay (SPDT 5V coil)	Sec 6	High	PLANNED	Replaces DPDT manual switch; ESP32 GPIO 5 → 2N2222 → relay coil. Fully automatic source switching.
ESP32 Bluetooth A2DP stack migration	Sec 2	High	PLANNED	Moves BT receive to ESP32 native BT Classic. Frees Cubetek exclusively for PS2 optical passthrough.
Web UI (ESP32-hosted)	Sec 10	Medium	PLANNED	Browser UI at ESP32 IP: source override, EQ settings, streaming stats, volume display.
Tier 1 IIR EQ (5-band parametric)	Sec 10	Medium	PLANNED	Applied in ESP32 firmware on I2S audio buffer before PCM5102A. Configurable via Web UI.
Soft limiter (−3 dBFS)	Sec 10	Medium	PLANNED	Pre-DAC soft limiting in ESP32 firmware. Prevents clipping on high-level sources.
Woofer protection HPF (40 Hz)	Sec 10	High	PLANNED	2nd order IIR Butterworth always-active high-pass. Protects MHC-GX2000 from sub-bass overexcursion.
INMP441 I2S microphone node	Sec 11	Medium	PLANNED	INMP441 on ESP32 I2S1 (GPIO 14/15/32). Simultaneous with PCM5102A on I2S0.
ESP-SR wake-word detection	Sec 11	Medium	PLANNED	On-device wake-word on ESP32 using ESP-SR library. Triggers NAS AI pipeline via MQTT.

Feature	Section	Priority	Status	Notes
Raspberry Pi Zero 2W upgrade (optional)	Sec 10/11	Low	PLANNED	Replaces ESP32 if compute budget is exceeded by simultaneous WiFi+DSP+mic+wake-word.
Navidrome music server (NAS)	Sec 12	Medium	PLANNED	Lightweight Subsonic API music server. Squeezelite-ESP32 integration for WiFi streaming.
Jellyfin media server (NAS)	Sec 12	Low	PLANNED	Video + audio library. DLNA endpoint for non-Squeezelite clients.
Lyrion Music Server / Squeezelite backend	Sec 12	Medium	PLANNED	Multi-room sync backbone. ESP32 runs Squeezelite-ESP32 as LMS player endpoint.
PCM5102A 96 kHz upgrade	Sec 1	Low	PLANNED	Increase I2S sample rate. Requires DMA buffer re-tuning. Enables hi-res streaming from NAS.

8.3 v2.1 — Advanced Features (Roadmap)

Feature	Section	Priority	Status	Notes
CamillaDSP FIR room correction (NAS)	Sec 12	High	PLANNED	REW-generated FIR filters loaded into CamillaDSP pipeline on NAS. Requires UMIK-1 measurement (Sec 11).
UMIK-1 / UMM-6 room measurement	Sec 11	High	PLANNED	One-time acoustic measurement. Generates FIR filter for CamillaDSP. Repeat after room changes.
Minimum-phase FIR export from REW	Sec 12	High	PLANNED	20–40 ms total latency with minimum-phase FIR. Linear-phase export reserved for analysis only.
whisper.cpp speech recognition (NAS)	Sec 12	Medium	PLANNED	On-device STT from post-wake audio stream. No cloud API. 1–3 second transcription latency.
Ollama + Llama 3 8B command interpreter (NAS)	Sec 12	Medium	PLANNED	Structured JSON command output from voice transcript. System prompt constrains command grammar.
Piper TTS response (NAS)	Sec 12	Medium	PLANNED	On-device text-to-speech for voice command responses. Streamed to ESP32 WiFi path for speaker output.
Home Assistant integration	Sec 12	Medium	PLANNED	Source state MQTT, HA automation triggers, room scene control, audio notification injection.
Loudness compensation (ISO 226)	Sec 10	Low	PLANNED	Volume-dependent bass/treble boost using ISO 226:2003 equal-loudness contour data. 2 additional IIR biquad stages.

Feature	Section	Priority	Status	Notes
TPA3116D2 thermal protection relay	Sec 10	Low	PLANNED	NTC thermistor on TPA3116D2 case → ESP32 ADC → relay mute at > 70°C. Hardware placement TBD.
Cubetek GPIO mode control	Sec 2	Medium	PLANNED	ESP32 asserts GPIO to Cubetek mode select pin for automatic BT/PS2 switching without physical button.
Digital volume control (digital pot)	Sec 12	Low	PLANNED	Replaces manual TPA3116D2 potentiometer with DS1882 or MCP4131 digital potentiometer for software volume control from Web UI / voice commands / HA.
LM2596 → TPS62xxx efficiency upgrade	Sec 5	Low	PLANNED	90% efficiency vs 75%. Reduces NAS standby heat. Relevant if system moves to battery backup.

8.4 v3.0 — Long-Term Research Directions

Direction	Description	Dependency
Multi-room audio expansion	Second ESP32 + PCM5102A + TPA3116D2 node in another room. LMS synchronises playback to < 10 ms inter-room. CamillaDSP serves independent room correction profile per player.	Lyrion Music Server (v2.0) + CamillaDSP (v2.1) must be operational
Modular PCB platform	Custom PCB integrating ESP32, PCM5102A DAC, IIR DSP coprocessor (or RPi CM4 module), relay, power regulation, and microphone on a single board. Deployable as a self-contained smart audio node per room.	v2.x firmware stability; electrical design skills or PCB design service
Custom wireless audio transport	Research into low-latency wireless audio distribution for multi-room without Ethernet — WiSA-inspired 5 GHz or 6 GHz IEEE 802.11ax (Wi-Fi 6) with QoS prioritisation for < 5 ms wireless audio latency.	Advanced RF engineering; not critical path
CamillaDSP active crossover (subwoofer)	Add a separate subwoofer amplifier channel. CamillaDSP applies LR4 crossover — low frequencies routed to subwoofer, high frequencies to MHC-GX2000. Requires second amplifier output and subwoofer hardware.	CamillaDSP v2.1 operational; subwoofer hardware procurement
REW multi-point measurement average	Measure room response at 5–9 positions around the listening area and average the RIR before FIR filter generation. Produces a correction filter valid for a wider listening area rather than a single sweet spot.	UMIK-1 + REW v2.1 operational; time investment for multi-point measurement

Section 9 — References

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Supplement A — ADC Signal-Presence Threshold Derivation

This supplement provides the complete mathematical derivation of the signal-presence threshold used in the Intelligent Source Routing firmware (Section 6). In v1.0, the threshold was stated as "> 50 ADC counts RMS" without derivation. This supplement closes that gap.

A.1 System Parameters

Parameter	Value	Source
ESP32 ADC reference voltage	3.3V	ESP32 datasheet — ADC1 with VREF = 3.3V
ESP32 ADC resolution	12-bit (4096 counts)	ESP32 datasheet
Voltage per count	$3.3\text{V} / 4096 = 0.806\text{ mV/count}$	Derived
Voltage divider ratio	$3.3\text{ k}\Omega / (10\text{ k}\Omega + 3.3\text{ k}\Omega) = 0.248$	Resistor values — Section 6.3.1
PCM5102A output (active)	2.1V RMS nominal	PCM5102A datasheet
Optical DAC output (active)	2.0V RMS nominal	Optical DAC specification
Thermal noise floor at ADC input	~1–2 mV RMS (typical)	Measured — oscilloscope at ADC pin with no source

A.2 Active Source Voltage at ADC Pin

For a PCM5102A output of 2.1V RMS sinusoidal audio at the sensing point:

Calculation Step	Formula	Result
Voltage at ADC pin (RMS)	$2.1\text{V} \times 0.248$ (divider ratio)	$\approx 521\text{ mV RMS}$
Peak voltage at ADC pin	$521\text{ mV} \times \sqrt{2}$	$\approx 737\text{ mV peak}$
ADC count at active RMS level	$521\text{ mV} / 0.806\text{ mV per count}$	$\approx 646\text{ ADC counts RMS}$

At an active source signal, the ADC reads approximately 646 counts RMS. The noise floor at the ADC input is approximately 2–3 ADC counts with no source connected.

A.3 Threshold Selection

Threshold	ADC Counts	Voltage at ADC Pin	Interpretation
Noise floor	2–3 counts	1.6–2.4 mV	No source; thermal noise only

Threshold	ADC Counts	Voltage at ADC Pin	Interpretation
Silent buffer (ESP32 idle output)	5–10 counts	4–8 mV	Source connected but not playing
Selected threshold	50 counts	40 mV	~12.5 dB above noise floor; ~22 dB below active level
Active source (nominal)	646 counts	521 mV RMS	Normal playback level

A threshold of 50 ADC counts (≈ 40 mV at the ADC pin) provides a 25:1 ratio above the noise floor and a 13:1 ratio below the active signal level. This gives robust separation with no risk of false detection from thermal noise, and reliable detection at volume levels as low as approximately -22 dBFS relative to full-scale PCM5102A output.

A.4 Minimum Detectable Volume Level

The threshold of 50 ADC counts corresponds to 40 mV at the ADC pin. The divider ratio back-calculates to a source RCA output of $40\text{ mV} / 0.248 = 161\text{ mV RMS}$ minimum detectable. This represents approximately -22 dBFS at the PCM5102A full-scale output of 2.1V RMS. Any playback at volume levels above approximately 1% of maximum will be reliably detected.

A.5 Summary

Parameter	Value
Voltage divider ratio	0.248 (3.3 k Ω lower / 13.3 k Ω total)
ADC resolution (mV/count)	0.806 mV/count
Threshold (ADC counts)	50 counts
Threshold (ADC pin voltage)	≈ 40 mV RMS
Threshold (source RCA voltage)	≈ 161 mV RMS
Threshold (relative to full-scale)	≈ -22 dBFS
Noise floor	2–3 ADC counts (< 2.5 mV at pin)
Threshold / noise margin	$\sim 25:1$ ($\approx +28$ dB above noise)
Signal / threshold margin	$\sim 13:1$ ($\approx +22$ dB above threshold at nominal level)

Supplement B — System Block Diagrams

This supplement provides textual block diagram descriptions for each subsystem. Full schematic drawings are maintained as companion CAD files (KiCad format) in the project repository. The diagrams below use ASCII-style notation to represent signal flow and component interconnects within this document.

B.1 Complete System Signal Flow

Domain	Signal Chain (left to right)	Isolation Type
WiFi Path	WiFi Network → ESP32 (802.11 RX) → I2S Bus → PCM5102A DAC → 10 kΩ series R → DPDT Switch Position A	Digital isolation (no copper to source)
BT Path	BT Device → Cubetek (A2DP RX) → Toslink → Optical DAC → 10 kΩ series R → DPDT Switch Position B	Galvanic isolation via Toslink
PS2 Path	PS2 Optical Out → Toslink → Cubetek (passthrough) → Toslink → Optical DAC → 10 kΩ series R → DPDT Switch Position B	Galvanic isolation via Toslink
Output Path	DPDT Switch → TPA3116D2 Input (RCA) → Class-D Amp → Speaker Terminals → Sony MHC-GX2000	N/A — fully analog final stage
Control Path	ESP32 ADC (GPIO 34/35) ← 10 kΩ + 3.3 kΩ dividers ← RCA sensing taps (pre-DPDT)	Passive tap only — no drive capability

B.2 Power Distribution Diagram

Node	From	To	Regulation	AWG
AC Mains	Wall outlet	19V Laptop Adapter	External (adapter)	N/A
19V Barrel Jack	Laptop Adapter	DC Power Splitter	None — direct	18 AWG
19V Muscle Rail	DC Splitter Output 1	TPA3116D2 VCC	None — unregulated direct	18 AWG
19V → LM2596 Input	DC Splitter Output 2	LM2596 Vin	None	20 AWG
5V Brain Rail	LM2596 Vout	ESP32 VIN, Optical DAC, Cubetek 5V	LM2596 buck regulated	22 AWG
3.3V Logic Rail	ESP32 onboard LDO	PCM5102A VCC (via ferrite bead)	ESP32 internal 3.3V LDO	24 AWG
Star GND Node	DC Barrel Jack GND	All component GND returns	0V reference — single star point	18–24 AWG (see Sec 5.4)

B.3 Source Routing Control Diagram

ESP32 Pin	Direction	Connected To	Signal Description
GPIO 34 (ADC)	Input only	10 k Ω + 3.3 k Ω divider tap from PCM5102A RCA out	WiFi path signal-presence voltage (0–3.3V). Threshold: 50 counts $\approx$ 40 mV — Supplement A.
GPIO 35 (ADC)	Input only	10 k Ω + 3.3 k Ω divider tap from Optical DAC RCA out	Multimedia path signal-presence voltage (0–3.3V). Same threshold.
GPIO 2 (Output)	Output	220 Ω $\rightarrow$ Blue LED $\rightarrow$ GND	HIGH when WiFi path is selected as active source.
GPIO 4 (Output)	Output	220 Ω $\rightarrow$ Green LED $\rightarrow$ GND	HIGH when Multimedia path is selected as active source.
GPIO 5 (Output, v2.0)	Output	1 k Ω $\rightarrow$ 2N2222 base; 2N2222 collector $\rightarrow$ relay coil; 1N4007 flyback across coil	HIGH drives relay to Optical DAC position (Multimedia). LOW = WiFi position.
GPIO 27 (Output)	Output	PCM5102A XSMT Pin 17	HIGH = play. LOW = soft mute DAC output during source switch transitions.

Supplement C — 19V Adapter Worst-Case Power Budget

This supplement derives the minimum required current rating for the 19V DC laptop adapter. The v1.0 TDD specified a '3A minimum' adapter without providing a derivation. This supplement demonstrates that 3A is insufficient under worst-case simultaneous load conditions, and establishes the corrected minimum specification.

C.1 System Load Summary at 19V Rail

Load	19V Rail Draw (Worst Case)	Derivation
TPA3116D2 amplifier (peak transient, 6Ω)	~3,000 mA	Datasheet max; bass transient peak at 19V into 6Ω: $P = V^2/2R = 361/12 \approx 30W$ per channel; $I = P/V = 30/19 \approx 1.6A$ per channel $\times 2 = 3.2A$ peak
LM2596 brain rail (19V input)	~200 mA at 19V input	Brain rail: 760 mA at 5V output; efficiency ~80%; $P_{in} = P_{out}/\eta = 5 \times 0.76/0.8 = 4.75W$; $I = 4.75W/19V \approx 250$ mA
Total worst-case at 19V rail	~3,250 mA	TPA3116D2 peak + LM2596 input current simultaneously

C.2 Adapter Specification Derivation

Requirement	Value	Rationale
Worst-case 19V current draw	3,250 mA	See C.1 derivation above
Safety margin factor	1.25×	Standard engineering margin for adapter derating over temperature
Required adapter current	$3,250 \text{ mA} \times 1.25 = 4,063 \text{ mA}$	Round up to next standard size
Minimum adapter specification	4A / 76W at 19V	Standard laptop adapter rating; 65W is marginal — use 90W for margin
Recommended adapter specification	4.7A / 90W at 19V	Common Lenovo/HP compatible adapter; provides meaningful headroom

C.3 Why v1.0 3A Specification Was Insufficient

The v1.0 specification of 3A minimum was likely based on the TPA3116D2 datasheet's typical average current at moderate listening levels (~1.5A) plus the brain rail draw (~250 mA), totalling approximately 1.75A — well within 3A. However, at peak transient bass output into 6Ω at 19V, the amplifier can draw up to 3.0A alone. Adding the 250 mA brain rail input current brings total peak draw to 3.25A, which exceeds a 3A adapter's rated output.

Consequence: a 3A adapter operating at 3.25A will engage its overcurrent protection, causing the supply voltage to drop or the adapter to shut down. This manifests as audio dropout, ESP32 reset, or system brownout during loud bass passages. A 4A minimum adapter eliminates this failure mode.

C.4 Corrected Specification

Parameter	v1.0 Specification	v1.1 Corrected Specification
Minimum adapter current	3A	4A (minimum) / 4.7A (recommended)
Minimum adapter wattage	57W	76W (minimum) / 90W (recommended)
Connector	5.5mm × 2.1mm DC barrel, centre positive	5.5mm × 2.1mm DC barrel, centre positive (unchanged)
Voltage	19V DC	19V DC (unchanged)

Supplement D — Contention State Full Exit Logic

This supplement defines the complete state machine exit logic for the CONTENTION state in the Intelligent Source Routing engine. In v1.0, the CONTENTION state held the current source and started a 3-second timer, but the exit conditions when both sources remained simultaneously active after the timer expired were not defined. This supplement closes that gap.

D.1 CONTENTION Scenarios

Scenario	Trigger	Duration	Appropriate Response
Brief simultaneous onset	Both sources become active within the same 500 ms activation window	< 3 seconds	Hold current source; one source will naturally become dominant
Game start during music	PS2 game audio activates while WiFi music is playing	Indefinite — both genuinely playing simultaneously	Apply precedence: WiFi (primary IoT path) wins; switch to WiFi after 3s
Notification over PS2	WiFi notification sound fires while playing PS2 game	< 3 seconds (notification is short)	Hold PS2 (current source); notification ends; PS2 resumes
True sustained contention	Both paths deliver continuous audio > 3 seconds	Indefinite	Apply precedence: WiFi > Multimedia; log CONTENTION_RESOLVE event

D.2 Updated State Machine — CONTENTION Branch

State	Entry Condition	Timer	Exit Condition	Next State	Action
CONTENTION	Both sources detected ACTIVE simultaneously	Start 3s timer	One source goes SILENT before timer expires	Return to previous single-source state	No switch — hold current source
CONTENTION	Both sources detected ACTIVE simultaneously	3s timer running	Timer expires; one source is now SILENT	Switch to remaining ACTIVE source	Switch relay/LED to newly dominant source
CONTENTION_RESOLVE	3s timer expires; BOTH sources still ACTIVE	None — immediate	Apply precedence : WiFi > Multimedia	WIFI_ACTIVE	Switch to WiFi; assert blue LED; log

State	Entry Condition	Timer	Exit Condition	Next State	Action
					resolution event
CONTENTION_RESOLVE	After precedence switch, Multimedia goes SILENT	None	Multimedia silent > 3s deactivation hold-off	WIFI_ACTIVE (stable)	No further action — stable
CONTENTION_RESOLVE	After precedence switch, WiFi goes SILENT	Deactivation 3s timer	WiFi silent for 3s; Multimedia still active	MULTIMEDIA_ACTIVE	Switch to Multimedia after WiFi deactivates

D.3 Precedence Rationale

WiFi is assigned higher precedence than Multimedia (BT + PS2) for the following reasons:

- The WiFi path is the primary IoT streaming endpoint — the feature that distinguishes A.U.R.A. from a passive audio switcher. Interrupting it in favour of legacy content would undermine the system's primary value proposition.
- In practice, WiFi audio is typically active because the user intentionally initiated a stream. PS2 audio can be active passively (menu loops, background music) even when the user is not actively engaged.
- In a future multi-source scenario where the user is simultaneously streaming music via WiFi and playing a PS2 game, the WiFi stream represents the intentional audio task.

Note: The precedence rule is a configurable parameter. A future Web UI (v2.0 roadmap) will allow the user to invert precedence or disable the rule in favour of manual override.

D.4 Updated Firmware Parameters for CONTENTION_RESOLVE

Parameter	Value	Rationale
Contention detection threshold	Both channels > 50 ADC counts simultaneously	Same threshold as single-source detection
Contention hold timer	3,000 ms	Tolerates brief mutual activity (notifications, fade-ins)
Precedence rule	WiFi > Multimedia	WiFi is primary IoT path — see D.3
Post-resolve deactivation hold	3,000 ms	Same as standard deactivation hold-off — prevents re-contention flutter
Resolution event logging	Serial: 'CONTENTION_RESOLVE: switched to WiFi at T=Xms'	Debug visibility for bench testing
User notification	Both LEDs blink during CONTENTION; blue LED	Visual feedback distinguishes contention from stable state

Parameter	Value	Rationale
	solid during CONTENTION_RESOLVE	

Supplement E — Output Impedance Protection Resistors

This supplement provides the engineering justification and implementation details for the 10 kΩ series resistors added between each DAC output and the DPDT switch in v1.1. The v1.0 design noted this as a limitation ('add 10 kΩ series resistors on each RCA input to the switch') but did not implement or document them.

E.1 The Problem: Mutual DAC Loading

In the v1.0 design, the PCM5102A RCA output and the Optical DAC RCA output both connect to the DPDT switch through passive wiring with no series isolation. When the DPDT switch is in Position A (WiFi active), the Optical DAC is electrically connected to the shared wiring node but is not selected. The Optical DAC's output stage presents a low-impedance path to AC ground — typically 100–600 Ω for line-level DAC outputs.

This creates a voltage divider between the active source (PCM5102A, ~100 Ω output impedance) and the idle source (Optical DAC, ~600 Ω output impedance). The combined load impedance seen by the PCM5102A becomes the parallel combination of the DPDT switch's downstream impedance (~10 kΩ TPA3116D2 input) and the idle Optical DAC output impedance (~600 Ω):

Calculation	Formula	Result
Parallel load impedance (no series R)	$10\text{ k}\Omega \parallel 600\text{ }\Omega = (10,000 \times 600) / (10,000 + 600)$	$\approx 566\text{ }\Omega$ effective load on active DAC
Voltage attenuation (no series R)	$566\text{ }\Omega / (100\text{ }\Omega + 566\text{ }\Omega)$	≈ 0.85 — 15% signal loss
PCM5102A output (2.1V RMS nominal)	$2.1\text{V} \times 0.85$	$\approx 1.79\text{V}$ RMS at amplifier input

In addition to signal loss, the idle DAC's non-linear output stage can introduce harmonic distortion products into the active signal path through the shared node.

E.2 The Solution: 10 kΩ Series Resistors

Adding a 10 kΩ series resistor in the RCA line between each DAC output and the DPDT switch isolates the two source impedances. When Position A is selected:

Calculation	Formula	Result
Idle Optical DAC load (via 10 kΩ series R)	$10\text{ k}\Omega \text{ series R} + 600\text{ }\Omega \text{ output Z} = 10.6\text{ k}\Omega \text{ to AC}$	Negligible load on active path
Combined load on active source	$10\text{ k}\Omega \text{ (TPA3116D2)} \parallel 10.6\text{ k}\Omega \text{ (idle path via series R)}$	$\approx 5.15\text{ k}\Omega$ — well within PCM5102A drive capability
Voltage divider with active series R	$5.15\text{ k}\Omega / (100\text{ }\Omega \text{ source Z} + 10\text{ k}\Omega \text{ active series R} + 5.15\text{ k}\Omega)$	≈ 0.337 — 6 dB attenuation
Active DAC output at amplifier	$2.1\text{V} \times 0.337 = 0.71\text{V RMS at TPA3116D2 input}$	Volume control adjusted to compensate — see E.3

E.3 Volume Compensation

The 6 dB attenuation introduced by the 10 kΩ series resistors reduces the TPA3116D2 input from 2.1V to ~0.71V RMS. Since the TPA3116D2 is a fixed-gain Class-D amplifier with a pre-gain volume potentiometer, this attenuation is compensated by advancing the volume control by approximately one-third of its rotation range (from the position that previously produced the desired loudness). No amplifier gain modification is required. This is an acceptable tradeoff: the idle-path noise isolation benefit significantly outweighs the minor volume control recalibration.

E.4 Component Specification and Placement

Parameter	Specification
Resistor value	10 kΩ ± 1%
Power rating	0.25W (125 mW — well within rated power at line level)
Quantity	4 total (L + R channels × 2 DAC paths)
Package	Through-hole 1/4W metal film (low noise)
Placement	Inline on each RCA hot wire, between DAC output connector and DPDT switch input lug
Grounding	No change — signal ground via RCA shield connects directly to star GND node

E.5 Before / After Comparison

Parameter	v1.0 (No Series R)	v1.1 (10 kΩ Series R)
Idle DAC mutual loading	~566 Ω — significant	~10.6 kΩ — negligible
Signal attenuation from loading	~15%	< 0.5%
Harmonic distortion from idle DAC	Present (idle DAC non-linearity)	Eliminated — isolated by 10 kΩ
Active signal at TPA3116D2 input	~1.79V RMS	~0.71V RMS (compensated by volume control)
Volume control recalibration needed	None	Advance by ~1/3 travel
Components added	None	4 × 10 kΩ 0.25W resistors

Supplement F — v1.0 Gap Closure Tracking Table

This supplement provides a consolidated audit of all gaps identified in the v1.0 Technical Design Document and their resolution status in v1.1.

Gap Identified	Severity	Resolution in v1.1	Location
ADC threshold (50 counts) stated without derivation	<i>Medium — academic rigour gap</i>	CLOSED	Full voltage divider and count calculation in Supplement A
No schematic or wiring diagram	<i>Medium — documentation gap</i>	PARTIALLY CLOSED	Textual block diagrams added in Supplement B; KiCad CAD files maintained as companion files
19V adapter minimum specified as 3A without derivation; insufficient at peak load	<i>High — hardware risk</i>	CLOSED	Full worst-case current budget in Supplement C; minimum corrected to 4A
CONTENTION state has no defined exit for sustained simultaneous activity	<i>High — firmware correctness gap</i>	CLOSED	Full CONTENTION_RESOLVE state and precedence logic in Supplement D; state machine updated in Section 6
10 k Ω series output resistors mentioned in limitations but not implemented	<i>Medium — implementation gap</i>	CLOSED	Resistors added to all BOMs and wiring tables; full impedance derivation in Supplement E
Academic terminology tables restate definitions already in prose	<i>Low — style observation</i>	ACKNOWLEDGED	Not changed — terminology tables are a submission requirement; IEEE/IEC citations added where applicable

Section 10 — DSP & Audio Processing Subsystem

10.1 Overview

The DSP & Audio Processing Subsystem introduces real-time digital signal processing into the A.U.R.A. signal chain. In v1.x, audio travels from source to amplifier with no correction, no equalisation, and no room compensation. The Sony MHC-GX2000 speakers have a non-flat frequency response, and the room's geometry introduces reflections, standing waves, and comb-filtering that colour the perceived sound. The DSP subsystem corrects all of these in the digital domain before the signal reaches the TPA3116D2 amplifier.

Processing is split across two tiers: lightweight, latency-sensitive operations (volume normalisation, basic EQ, soft limiting) run on the ESP32 or a Raspberry Pi node; heavy operations (FIR room correction, multi-band crossover, delay alignment) run on the NAS via CamillaDSP, described in Section 12.

10.2 Design Philosophy

10.2.1 Why DSP Before Amplification (Not After)

DSP inserted before the amplifier operates on the line-level analog or digital signal at low power. It corrects deficiencies before they are amplified and delivered to the speaker. Correction after the speaker (acoustic treatment) is effective but expensive and non-adjustable. Correction in the amplifier output stage requires custom hardware. Pre-amplifier DSP is the most practical and reversible approach.

10.2.2 FIR vs IIR Filters

Two filter families are available for room correction and EQ:

- IIR (Infinite Impulse Response): Low computational cost, suitable for parametric EQ (biquad filters). Phase response is non-linear — each frequency band is delayed by a different amount, introducing phase shift that can blur transients. Used for simple EQ corrections on the ESP32/RPi tier.
- FIR (Finite Impulse Response): Higher computational cost, but phase-linear — all frequencies are delayed equally, preserving transient integrity. Required for precise room correction filters generated by REW (Room EQ Wizard). Run on the NAS tier via CamillaDSP.

10.2.3 Processing Architecture — Two-Tier

Tier	Platform	Operations	Latency Added	Status
Tier 1 — Edge	ESP32 / Raspberry Pi Zero 2W	Volume normalisation, soft limiter, basic 5-band IIR EQ, loudness compensation	< 5 ms	Planned v2.0
Tier 2 — NAS	NAS (x86/ARM SBC) via CamillaDSP	Full FIR room correction (1024–65536 tap filters), multi-band crossover, delay alignment, dynamic range compression	10–30 ms (buffered)	Planned v2.1

10.3 Hardware Architecture

10.3.1 Tier 1 — ESP32 / RPi DSP Signal Chain

Stage	Component	Operation	Interface
Digital input	ESP32 I2S receive (WiFi path)	PCM audio buffer received over WiFi	I2S DMA
DSP processing	ESP32 firmware (or RPi Zero 2W)	5-band parametric IIR EQ, soft limiter, loudness curve	In-memory DSP on audio buffer
Volume normalisation	ESP32 firmware	RMS-based gain riding to maintain consistent perceived loudness across sources	Applied to I2S output buffer
DAC output	PCM5102A (existing)	Processed PCM → analog line level	I2S → PCM5102A
Analog out	RCA → 10 kΩ series R → DPDT	Line level to amplifier (unchanged from v1.1)	Analog

10.3.2 Tier 1 — IIR Parametric EQ Specification

Band	Centre Frequency	Type	Default Q	Default Gain	Purpose
Band 1	60 Hz	Low shelf	0.7	0 dB (adjustable)	Bass shelf — compensate MHC-GX2000 bass rolloff
Band 2	200 Hz	Peak/Notch	1.4	0 dB (adjustable)	Mud reduction — common room resonance zone
Band 3	1 kHz	Peak/Notch	1.0	0 dB (adjustable)	Midrange balance — vocal clarity
Band 4	5 kHz	Peak/Notch	1.2	0 dB (adjustable)	Presence / detail boost
Band 5	12 kHz	High shelf	0.7	0 dB (adjustable)	Air shelf — compensate tweeter rolloff

All EQ parameters are user-configurable via the v2.0 Web UI (ESP32-hosted) or via the NAS CamillaDSP interface. Default gains are 0 dB (flat) at first boot. Recommended starting values for the Sony MHC-GX2000 are derived from the REW measurement process described in Section 11 (Microphone Subsystem).

10.3.3 Soft Limiter Specification

Parameter	Value	Rationale
Threshold	−3 dBFS	3 dB headroom below digital full-scale; prevents inter-sample peaks causing clipping at DAC output

Parameter	Value	Rationale
Knee	2 dB soft knee	Gradual onset prevents audible pumping on transients
Attack time	0.5 ms	Fast enough to catch transients before they clip
Release time	50 ms	Slow enough to avoid gain pumping on sustained signals
Gain reduction range	0 to -12 dB	Beyond 12 dB reduction, signal is too distorted to recover cleanly — hard clip engaged

10.3.4 Loudness Compensation

At low listening volumes, human hearing sensitivity drops disproportionately in the bass and treble regions (Fletcher-Munson curves / ISO 226:2003 equal-loudness contours). Loudness compensation applies a volume-dependent boost to bass and treble to maintain perceived spectral balance at low volumes. The correction curve is applied as two additional IIR biquad stages:

- Bass boost (60–200 Hz): up to +6 dB at minimum volume, tapering to 0 dB at 60% volume setting.
- Treble boost (8–15 kHz): up to +3 dB at minimum volume, tapering to 0 dB at 60% volume setting.
- Boost curves are derived from ISO 226:2003 equal-loudness contour data, scaled to the TPA3116D2 volume control position reported via the Web UI.

10.4 Tier 2 — NAS CamillaDSP (Full Detail in Section 12)

Full FIR-based room correction, multi-band crossover tuning, and delay correction are handled by CamillaDSP running on the NAS. This tier is described in detail in Section 12 (NAS Integration Subsystem). The DSP pipeline between the ESP32/RPi and CamillaDSP is:

- ESP32/RPi streams processed PCM audio to the NAS over the local network via a low-latency ALSA loopback or Squeezelite pipeline.
- CamillaDSP applies FIR room correction filters (generated by REW from microphone measurements — Section 11), crossover tuning, and delay alignment.
- CamillaDSP outputs the corrected audio stream back to the network, where it is delivered to the PCM5102A for final DAC conversion.

10.5 Speaker Protection

Speaker protection is implemented as a DSP-level safety layer. The Sony MHC-GX2000 woofer is a vented (bass-reflex) design with an approximate low-frequency rolloff below 50 Hz at -6 dB/octave below the port tuning frequency. Below the port frequency, cone excursion increases dramatically with decreasing frequency, risking voice coil damage at high volumes.

Protection Feature	Implementation	Parameters	Stage
High-pass filter (woofer protection)	2nd order Butterworth IIR HPF	Fc = 40 Hz, -12 dB/octave below 40 Hz	Tier 1 (ESP32/RPi) — always active

Protection Feature	Implementation	Parameters	Stage
Soft clip limiter	Brickwall soft limiter at -3 dBFS	See 10.3.3	Tier 1 — always active
FIR bass shelf correction	REW-generated FIR filter targeting MHC-GX2000 bass response	Filter taps: 4096–16384, FS = 44.1/48 kHz	Tier 2 (NAS CamillaDSP)
Thermal protection relay (future)	ESP32 monitors TPA3116D2 temperature via NTC thermistor	Relay mute triggered at > 70°C case temp	Planned v2.1 — hardware not yet specified

10.6 Bill of Materials — DSP Subsystem

Qty	Component	Value / Part	Purpose	Status	Supplier
1	Raspberry Pi Zero 2W (optional upgrade)	RPi Zero 2W — 512 MB RAM, quad-core 1 GHz	Replaces ESP32 as Tier 1 DSP node; higher compute for IIR EQ and Spotify Connect	Planned v2.0	Official RPi / Robu.in
1	PCM5102A (existing)	32-bit / 384 kHz	DAC for Tier 1 processed output — unchanged from v1.1	Existing	—
1	NTC Thermistor	10 kΩ NTC (B = 3950)	TPA3116D2 case temperature monitoring for thermal protection	Planned v2.1	Amazon / Robu.in
1	Resistor (thermistor divider)	10 kΩ 1% fixed	Voltage divider with NTC for ESP32 ADC temperature sensing	Planned v2.1	Amazon
—	CamillaDSP software	Open source — v2.0.x (Linux)	NAS-hosted FIR room correction and crossover engine	Planned v2.1	github.com/HEnquist/camilladsp
—	REW (Room EQ Wizard)	Free — Windows/macOS/Linux	Acoustic measurement and FIR filter generation	Planned v2.1	roomeqwizard.com

Qty	Component	Value / Part	Purpose	Status	Supplier
			from microphone data		

10.7 Engineering Assessment — DSP Subsystem

Strengths

- Two-tier architecture keeps always-on safety features (HPF, soft limiter) at the ESP32/RPi edge with sub-5ms latency, while heavy FIR correction runs on the NAS where compute is not constrained.
- IIR biquad filters (Tier 1) are computationally trivial for the ESP32 — a single biquad costs approximately 5 multiply-accumulate operations per sample, giving 5 bands × 44,100 samples/sec = 220,500 MAC/sec, well within ESP32 capability.
- FIR room correction (Tier 2 / NAS) is phase-linear — preserves transient attack and decay which IIR correction cannot.
- Speaker protection HPF at 40 Hz is a hardware-independent safety net regardless of source or volume setting.
- All EQ parameters are remotely configurable via Web UI — no physical access needed.

Limitations

- Tier 2 NAS DSP introduces 10–30 ms of additional latency. Acceptable for music; may cause lip-sync issues for TV/PS2 video content. Mitigation: Tier 2 DSP can be bypassed for the Optical DAC path (PS2/TV), with only Tier 1 edge processing applied.
- ESP32 compute budget is tight when running simultaneous WiFi streaming, I2S DMA, signal-presence ADC, and DSP. Raspberry Pi Zero 2W (v2.0 upgrade) eliminates this constraint.
- Thermal protection relay is not yet specified at hardware level — NTC thermistor placement and GPIO wiring are pending physical access to TPA3116D2 board for mounting point selection.

10.8 Academic Terminology Reference — DSP Subsystem

Technique	Formal Description
Biquad IIR Filter	Second-order Infinite Impulse Response filter implementing the Direct Form II transposed structure: $H(z) = (b_0 + b_1z^{-1} + b_2z^{-2}) / (1 + a_1z^{-1} + a_2z^{-2})$. Each biquad stage provides one resonance peak, shelf, or band-pass characteristic.
FIR Filter	Finite Impulse Response filter: output is a weighted sum of current and past input samples only. Always stable; phase response is linear when filter coefficients are symmetric (linear phase FIR). Computationally expensive: an N-tap FIR requires N multiply-accumulate operations per sample.
Room Correction	Measurement and inversion of the room's acoustic transfer function. A measurement microphone captures the room impulse response (RIR); minimum-phase or linear-phase inverse filters are computed from the RIR

Technique	Formal Description
	and applied in the DSP chain to flatten the perceived frequency response at the listening position.
Equal-Loudness Contours (ISO 226)	Standardised curves describing the sound pressure level required at each frequency to produce the same perceived loudness. At low volumes, bass and treble require disproportionately higher SPL — loudness compensation pre-emphasises these regions accordingly.
Fletcher-Munson Effect	Perceptual phenomenon (confirmed by ISO 226:2003) where human hearing sensitivity is non-uniform across frequency at different SPLs. Bass sensitivity decreases sharply at low listening volumes, justifying loudness compensation in consumer audio systems.
Soft Limiter / Look-Ahead Limiter	A gain processor that monitors the audio signal for peaks exceeding a threshold and reduces gain smoothly (soft knee) to prevent clipping. Attack time determines how quickly gain is reduced; release time determines recovery speed. Prevents inter-sample clipping that occurs when bandwidth-limited DAC reconstruction exceeds 0 dBFS between sample points.

Section 11 — Microphone & Room Correction Subsystem

11.1 Overview

The Microphone Subsystem serves two distinct roles in A.U.R.A.: acoustic measurement for room correction calibration (a one-time or periodic setup procedure), and a persistent microphone node for voice control, intercom, and local AI assistant interaction. These roles share the same physical hardware but operate in different modes with different firmware contexts.

11.2 Design Philosophy

11.2.1 Why a Dedicated Measurement Microphone

Consumer microphones (phone mics, webcam mics) have non-flat frequency responses with significant colouration above 5 kHz and below 100 Hz. Room correction filters generated from coloured microphone measurements contain the microphone's own response errors superimposed on the room's response — the filters then overcorrect at microphone-artefact frequencies. A calibrated measurement microphone with a known deviation file (supplied by the manufacturer) allows REW and CamillaDSP to subtract the microphone's own response, leaving only the room's contribution in the correction filter.

11.2.2 Measurement Mode vs Persistent Mode

Mode	When Active	Hardware Used	Output	Latency Requirement
Measurement Mode	One-time room calibration; periodic re-measurement after room changes	Measurement mic + USB audio interface or I2S mic module → NAS → REW	Room Impulse Response (RIR) → REW FIR filter export → CamillaDSP filter load	None — offline measurement
Persistent Mode	Always-on during system operation	I2S MEMS microphone (INMP441 or SPH0645) → ESP32 or RPi	Wake-word detection, voice commands, intercom audio stream, room presence detection	< 100 ms end-to-end for voice response

11.3 Hardware Architecture — Measurement Mode

11.3.1 Recommended Measurement Microphone

Parameter	Specification	Notes
Recommended model	Dayton Audio UMM-6 or UMIK-1	Both include manufacturer-supplied individual calibration file (USB). UMIK-1 is plug-and-play USB; UMM-6 requires USB audio interface.
Frequency range	20 Hz – 20 kHz ± 1 dB (calibrated)	Raw capsule ± 3 dB; calibration file corrects to ± 1 dB

Parameter	Specification	Notes
Polar pattern	Omnidirectional	Required for room impulse response measurement — captures all reflections equally
Sensitivity	−37 dBV/Pa typical (MiniDSP UMIK-1)	Adequate for REW sweep levels at normal listening distances
Interface	USB (Class-compliant audio device)	Plug-and-play on NAS Linux without driver installation
Calibration file format	.cal (REW-compatible)	Loaded into REW to subtract microphone response from measured room response

11.3.2 Measurement Signal Chain

Stage	Component	Function
Stimulus generation	REW on NAS — log-swept sine (20 Hz to 20 kHz)	Generates calibrated test signal played through TPA3116D2 → Sony speakers
Acoustic capture	Measurement microphone at listening position	Captures room impulse response including all reflections and modes
ADC conversion	USB audio interface (or UMIK-1 direct)	Digitises microphone signal at 48 kHz / 24-bit minimum
RIR computation	REW software on NAS	Deconvolves captured signal with stimulus to extract Room Impulse Response
Filter export	REW → Export FIR filter as WAV (minimum-phase or linear-phase)	32,768 or 65,536 tap filter at 48 kHz sampling rate
Filter load	CamillaDSP configuration on NAS	FIR filter WAV loaded as convolver in CamillaDSP pipeline

11.3.3 REW Measurement Procedure

The standard room correction measurement procedure for A.U.R.A. is as follows. This procedure should be repeated after any significant change in room furnishings, speaker placement, or listening position:

- Step 1: Place the measurement microphone at ear height at the primary listening position. Microphone should be on a stand — hand-holding introduces body reflection artefacts.
- Step 2: Connect the measurement microphone to the NAS via USB. Verify REW detects it as input device and loads the calibration file.
- Step 3: Set A.U.R.A. to play audio through the WiFi path (PCM5102A → TPA3116D2 → Sony speakers). Verify the REW output is routed through A.U.R.A.'s digital chain.
- Step 4: In REW, run a log-swept sine measurement at −12 dBFS sweep level. This is the Room Impulse Response (RIR) capture.
- Step 5: Apply REW's EQ suggestion filter — inspect for problematic notches below 80 Hz (likely standing wave modes). Apply smoothing (1/6 octave) to avoid over-correcting narrow nulls that move with head position.

- Step 6: Export the target EQ correction as a minimum-phase or linear-phase FIR filter WAV file at the CamillaDSP pipeline sample rate (44.1 or 48 kHz).
- Step 7: Load the exported filter into CamillaDSP as a Convolver block in the signal pipeline. Test at moderate volume and verify flat frequency response in REW measurement.
- Step 8: Save the CamillaDSP configuration. The filter is now active on all audio paths routed through the NAS DSP tier.

11.4 Hardware Architecture — Persistent Mode

11.4.1 MEMS I2S Microphone Selection

Parameter	INMP441	SPH0645LM4H	Recommendation
Interface	I2S (direct ESP32 compatible)	I2S (Adafruit breakout available)	Either — both I2S compatible with ESP32
SNR	61 dB A-weighted	65 dB A-weighted	SPH0645 marginally quieter — prefer for voice quality
Frequency response	60 Hz – 15 kHz ± 3 dB	100 Hz – 10 kHz flat; ± 3 dB to 15 kHz	INMP441 better low-frequency response
Operating voltage	1.8V – 3.3V	3.3V	Both compatible with ESP32 3.3V I2S pins
Package	LGA (SMD) — breakout board required	LGA (SMD) — Adafruit breakout available	Purchase breakout board for both
Power consumption	< 500 µA	< 650 µA	Both negligible on 5V brain rail

11.4.2 INMP441 I2S Pin Mapping — ESP32

Signal	ESP32 GPIO	INMP441 Pin	Notes
WS (Word Select / LRCLK)	GPIO 15	WS	Selects Left or Right channel — tie L/R pin to GND for left channel mono
SCK (Bit Clock)	GPIO 14	SCK	I2S bit clock from ESP32 I2S peripheral (I2S port 1 — separate from PCM5102A on I2S port 0)
SD (Data Out)	GPIO 32	SD	Serial PDM/I2S data from microphone to ESP32
VDD	3.3V	VDD	Logic and power supply
GND	Star GND node	GND	Connected to system star ground
L/R channel select	GND (tied)	L/R	GND = Left channel; 3.3V = Right channel. Use Left for mono voice capture.

Note: The ESP32 has two independent I2S peripherals (I2S0 and I2S1). PCM5102A uses I2S0 (GPIO 22/25/26). INMP441 uses I2S1 (GPIO 14/15/32). Both can operate simultaneously without conflict — I2S0 outputs audio to DAC while I2S1 captures microphone input.

11.4.3 Wake-Word Detection

Wake-word detection runs locally on the ESP32 or Raspberry Pi node using a lightweight neural network model. No audio is transmitted to an external cloud service — all detection is on-device, preserving privacy.

Parameter	Specification	Notes
Framework	ESP-SR (Espressif Speech Recognition) on ESP32	Espressif's on-device wake-word and speech recognition library. Alternatively: Porcupine (Picovoice) on RPi.
Wake word (default)	'Hey AURA' or custom model	ESP-SR allows custom wake word training via Espressif's model training pipeline
Detection latency	< 200 ms from word end to detection event	ESP-SR typical latency on ESP32 at 16 kHz input
False positive rate	< 1 per hour at typical room noise levels	ESP-SR specification at SNR > 10 dB
Audio capture rate	16 kHz / 16-bit mono	ESP-SR requirement; I2S1 configured for 16 kHz capture independent of PCM5102A 44.1 kHz output
Post-wake action	ESP32 asserts wake-event GPIO → NAS local AI pipeline (Section 12)	NAS handles speech recognition and command execution

11.5 Voice Control Command Set (Planned)

Voice Command	Action	Implemented By	Status
'Switch to WiFi'	Force source routing to WiFi/PCM5102A path	ESP32 relay GPIO	Planned v2.0
'Switch to Bluetooth'	Force source routing to Multimedia path	ESP32 relay GPIO	Planned v2.0
'Volume up / down'	Increment TPA3116D2 volume control via motor driver or digital pot	Hardware TBD — requires motor driver or digital potentiometer upgrade	Planned v2.1
'Play [playlist/album]'	Trigger Navidrome/Jellyfin playback on WiFi path	NAS local AI → Navidrome API	Planned v2.1
'What's playing'	Read current track metadata from Navidrome	NAS local AI → TTS → speaker	Planned v2.1
'Room off'	Trigger Home Assistant scene — lights off, system mute	NAS local AI → Home Assistant API	Planned v2.1

11.6 Intercom / Room Presence (Future)

The persistent microphone node also supports two future extensions:

- **Intercom:** Audio captured by the INMP441 can be streamed to a second A.U.R.A. node in another room over the local network using a low-latency audio streaming protocol (Opus codec over UDP, target < 50 ms). This enables room-to-room voice communication without a phone.
- **Room presence detection:** Continuous ambient audio analysis can detect whether a room is occupied (sound of movement, breathing, speech activity) without requiring an explicit wake word. Presence state is reported to Home Assistant for automation triggers (lights, HVAC, audio system standby).

11.7 Bill of Materials — Microphone Subsystem

Qty	Component	Value / Part	Purpose	Status	Supplier
1	Measurement microphone	MiniDSP UMIK-1 (USB, calibrated) or Dayton Audio UMM-6	Room impulse response measurement for REW/CamillaDSP correction	Planned v2.1	Amazon / Parts Express
1	MEMS I2S microphone	INMP441 breakout or SPH0645 (Adafruit)	Always-on voice capture, wake-word detection	Planned v2.0	Amazon / Adafruit
1	Microphone stand / clip	Desktop gooseneck or clip mount	Holds measurement mic at ear height during calibration	Planned v2.1	Amazon
2	Resistor	10 kΩ (I2S pull-up if needed)	SCK/WS line termination — check INMP441 breakout; may be on-board	Planned v2.0	Amazon
—	ESP-SR library	Espressif ESP-SR (open source)	On-device wake-word and speech recognition for ESP32	Planned v2.0	github.com/espressif/esp-sr
—	Picovoice Porcupine (alt)	Free tier available (RPi target)	Alternative wake-word engine if RPi replaces ESP32 in v2.0	Alternative	picovoice.ai

11.8 Engineering Assessment — Microphone Subsystem

Strengths

- Calibrated measurement microphone with manufacturer deviation file gives REW the data it needs to separate room response from microphone colouration — essential for accurate FIR filter generation.
- INMP441 I2S interface connects directly to ESP32 I2S1 — no ADC conversion, no analog noise floor, no level-matching required.
- Two independent I2S peripherals on ESP32 (I2S0 = output to PCM5102A, I2S1 = input from INMP441) allow simultaneous audio playback and voice capture without time-multiplexing or DMA conflicts.
- On-device wake-word detection (ESP-SR) means no audio is ever sent to an external cloud — fully privacy-preserving.

Limitations

- ESP32 running simultaneous WiFi streaming (I2S0 DMA), INMP441 capture (I2S1 DMA), signal-presence ADC, and ESP-SR wake-word inference will be compute-constrained. Raspberry Pi Zero 2W (v2.0) is the recommended upgrade to resolve this.
- INMP441 SNR of 61 dB is adequate for voice capture but not suitable as a measurement microphone — the noise floor would introduce errors in REW impulse response measurements at the quiet tail of the RIR. Always use the calibrated USB microphone for measurement mode.
- Wake-word false positive rate increases in rooms with television or music playback — threshold tuning may be required per room acoustic environment.

Section 12 — NAS Integration Subsystem

12.1 Overview

The NAS (Network Attached Storage) Integration Subsystem transforms the NAS from a passive file server into the central intelligence node of A.U.R.A. It hosts the media server stack (Jellyfin, Navidrome), the DSP processing engine (CamillaDSP), the local AI assistant stack (whisper.cpp, Ollama, Piper TTS), and the home automation integration layer (Home Assistant). All of these services run locally — no external cloud dependency.

The NAS communicates with the ESP32/RPi audio endpoint over the local network using established open protocols. It does not replace the ESP32's real-time audio functions; it augments them with services that require persistent storage, higher compute, and broad network connectivity.

12.2 NAS Hardware Specification

12.2.1 Minimum Viable NAS Specification

Parameter	Minimum Specification	Recommended Specification	Rationale
CPU	Intel Celeron J4125 / AMD Ryzen Embedded V1202B (x86-64)	Intel Core i3-N305 or Ryzen 5 5600G	CamillaDSP FIR convolution with 65k-tap filters at 48 kHz requires ~200 MFLOPS minimum. Whisper.cpp transcription requires AVX2 instruction set for acceptable inference speed.
RAM	8 GB DDR4	16 GB DDR4	Jellyfin transcoding (if needed) + CamillaDSP + Ollama LLM base model (7B parameter) requires > 8 GB. 16 GB allows 13B model.
Storage	2× 2 TB HDD (RAID 1) + 128 GB SSD (OS/apps)	2× 4 TB HDD (RAID 1) + 256 GB NVMe (OS/apps)	Music library: lossless FLAC at ~30 MB/album averages 1,000 albums per 30 GB. HDD for media; SSD/NVMe for OS, Docker, CamillaDSP filter files.
Network	1 GbE LAN	2.5 GbE LAN	Uncompressed 48 kHz / 32-bit stereo audio stream = 3 Mbps — well within 1 GbE. 2.5 GbE for multi-room future expansion.
OS	TrueNAS Scale (Linux/Kubernetes) or Debian 12	Debian 12 LTS (bare metal)	Docker-native for all services. TrueNAS Scale adds storage management UI. Bare Debian gives

Parameter	Minimum Specification	Recommended Specification	Rationale
			lowest overhead for CamillaDSP real-time audio.

12.3 Software Stack

12.3.1 Service Architecture

Service	Software	Version	Role	Interface	Status
Media server (video)	Jellyfin	10.x	Video and music library management, transcoding, web UI	HTTP API / DLNA / Chromecast	Planned v2.0
Music server	Navidrome	0.52+	Music-specific server with Subsonic API; lightweight; Spotify-Connect-like browsing	Subsonic API / OpenSubsonic	Planned v2.0
DSP engine	CamillaDSP	2.0+	FIR room correction, IIR EQ, crossover, multi-channel mixing pipeline	WebSocket API / YAML config	Planned v2.1
Squeezelite server	Lyrion Music Server (LMS)	8.x	Squeezelite-ESP32 backend — manages multi-room sync, protocol bridging	Squeezelite protocol / HTTP	Planned v2.0
Voice recognition	whisper.cpp	Latest stable	On-device speech-to-text from wake-word-triggered audio stream	C++ binary / REST wrapper	Planned v2.1
Local LLM	Ollama + Llama 3 8B	Latest stable	Natural language command interpretation from Whisper transcript	Ollama REST API	Planned v2.1
TTS output	Piper TTS	Latest stable	Text-to-speech response generation — low-latency, on-device	CLI / Python API	Planned v2.1
Home automation	Home Assistant	2024.x+	Smart-room control, automation triggers, device management	REST API / MQTT / Webhooks	Planned v2.1
Reverse proxy	Nginx or Caddy	Latest stable	Unified HTTPS access to all local services on port 443	HTTP/HTTPS	Planned v2.0

12.4 CamillaDSP Integration — Full Detail

12.4.1 CamillaDSP Pipeline Architecture

CamillaDSP is an open-source, ALSA-based audio processing pipeline that runs on the NAS as a real-time DSP engine. It receives audio from the network (via ALSA loopback, Squeezelite output, or direct

network pipe), applies a user-defined filter graph (YAML configuration), and outputs processed audio back to the network endpoint (ESP32/RPi PCM5102A chain).

Pipeline Stage	CamillaDSP Block Type	Configuration	Frequency Coverage
Input	ALSA capture device or network pipe (Squeezelite loopback)	48 kHz / 32-bit stereo	Full band
DC blocker	IIR High-pass filter — 1st order Butterworth	Fc = 5 Hz — removes DC offset and sub-infrasonic content	DC to 5 Hz (attenuated)
Woofer protection HPF	IIR — 2nd order Butterworth	Fc = 40 Hz — matches Tier 1 edge DSP protection (Section 10.5)	< 40 Hz attenuated
Room correction FIR	Convolver — FIR filter WAV file	32,768 to 65,536 taps at 48 kHz — generated by REW from UMIK-1 measurement (Section 11)	20 Hz – 20 kHz
Parametric EQ trim	IIR biquad cascade — up to 8 bands	Fine adjustment after FIR correction — corrects residual narrow-band peaks not captured in FIR	Targeted narrow bands
Crossover (future)	Linkwitz-Riley 4th order IIR (LR4)	Fc = 80–120 Hz — if separate subwoofer added in future	Sub / satellite split
Soft limiter	Gain + hard clip	Threshold: -1 dBFS — final protection before output	Full band
Output	ALSA playback device or network pipe to ESP32/RPi	48 kHz / 32-bit stereo	Full band

12.4.2 CamillaDSP YAML Configuration Structure

The CamillaDSP pipeline is defined in a YAML configuration file. The following describes the key structural elements for A.U.R.A.'s room correction pipeline:

- **devices:** defines the input (ALSA capture from Squeezelite loopback) and output (ALSA playback to network audio sink) devices, sample rate (48000), sample format (S32LE), and channel count (2).
- **filters:** named filter definitions — each filter specifies type (IIR Biquad, Conv/FIR, Gain), parameters (frequency, Q, gain for IIR; filename for FIR WAV), and sample rate.
- **pipeline:** ordered list of filter application steps. Each step references a named filter and specifies which audio channels it applies to (e.g., channel 0 = Left, channel 1 = Right).
- **mixers:** (future) allows channel routing and upmixing — used for crossover splitting to subwoofer output channel.
- The complete YAML configuration is maintained as a versioned file in the NAS repository alongside the REW-exported FIR filter WAV files.

12.4.3 Latency Budget — NAS DSP Tier

Latency Source	Typical Value	Notes
Network audio input buffer (ESP32 → NAS)	10–20 ms	UDP packet jitter buffer on NAS receive side
CamillaDSP processing buffer	5–10 ms	ALSA buffer size: 256–512 frames at 48 kHz
FIR convolver latency (linear-phase 32k taps)	~341 ms	Linear-phase FIR half-length delay: 32768/2 / 48000 Hz. Mitigated by using minimum-phase FIR export from REW (< 10 ms group delay).
Network audio output buffer (NAS → ESP32)	5–10 ms	Return path buffer
Total (minimum-phase FIR)	~20–40 ms	Acceptable for music; marginal for video. Bypass NAS DSP tier for PS2/TV Optical path.
Total (linear-phase FIR)	~360–380 ms	Unacceptable for video lip-sync. Use minimum-phase export for all operational configurations.

Recommendation: always export REW correction filters in minimum-phase mode for A.U.R.A. The phase response of minimum-phase FIR filters is non-linear but the group delay is < 10 ms — indistinguishable from a linear-phase FIR for music listening purposes. Reserve linear-phase filters for future measurement/analysis only.

12.5 Media Server Integration

12.5.1 Navidrome — Music Server

Parameter	Specification
Protocol	Subsonic API (OpenSubsonic extension) — compatible with DSub, Symfonium, Ultrasonic, and dozens of client apps
Library scan	Automatic on file change — monitors NAS music share directory
Supported formats	FLAC, ALAC, MP3, AAC, OGG, Opus, WAV — transcoding on-the-fly if client requests lower bitrate
Playback path	Navidrome → Subsonic API → Squeezelite-ESP32 on ESP32 → I2S → PCM5102A → TPA3116D2 → Sony speakers
Web UI	Built-in web player accessible at http://nas-ip:4533 — no app required for browser playback
Multi-user	Supports multiple user accounts — can share library with household members independently
Scrobbling	Last.fm / ListenBrainz compatible — listening history tracked if desired

12.5.2 Jellyfin — Video and Extended Media

Jellyfin provides video library management, episode tracking, and live TV (with tuner hardware) in addition to music. For A.U.R.A.'s audio-primary use case, Jellyfin's primary role is:

- Serving movie/TV audio to the speaker system when the source is a media player (not PS2/TV optical) — routed via DLNA or Chromecast to the ESP32 WiFi streaming path.
- Providing a unified interface for both music and video libraries accessible from any device on the local network.
- Optionally acting as a DLNA renderer target, so any DLNA controller (phone, smart TV app) can push audio to A.U.R.A. as a DLNA renderer endpoint.

12.6 Local AI Assistant Stack

12.6.1 Architecture Overview

The local AI assistant stack enables voice control of A.U.R.A. and smart-room functions without any cloud dependency. The pipeline is:

Stage	Software	Input	Output	Latency
Wake word detection	ESP-SR on ESP32 / Porcupine on RPi	Continuous INMP441 microphone audio stream (16 kHz / 16-bit)	Wake event GPIO pulse to NAS (via MQTT or GPIO)	< 200 ms from wake word end
Audio capture	ESP32/RPi streams 3–5 seconds of post-wake audio to NAS via UDP	Post-wake audio buffer	Raw PCM audio file or stream on NAS	Streaming — < 100 ms network latency
Speech-to-text	whisper.cpp on NAS (tiny or base model)	Raw PCM audio	Transcript text string	1–3 seconds (base model on modern x86 CPU)
Command interpretation	Ollama + Llama 3 8B on NAS	Transcript + system prompt defining command grammar	Structured JSON command object	1–5 seconds depending on prompt length
Command execution	Python dispatcher on NAS	JSON command object	API calls to Navidrome / CamillaDSP / Home Assistant / ESP32 relay GPIO	< 500 ms
TTS response	Piper TTS on NAS	Response text from LLM	WAV audio file / stream	< 1 second for short responses
Speaker playback	Response WAV streamed to ESP32 WiFi path → PCM5102A → speakers	WAV stream	Audio output through A.U.R.A. speakers	< 500 ms from TTS output

12.6.2 System Prompt Design for Command Grammar

The LLM is not used for general-purpose conversation — it is constrained by a structured system prompt that defines the available commands and their parameters. This reduces hallucination risk and ensures predictable JSON output. The system prompt structure:

- Defines the available command set (source switch, volume, playback control, home automation, query).
- Instructs the model to respond ONLY with a JSON object matching the command schema — no prose, no explanation.
- Includes examples of correct input/output pairs (few-shot prompting) for the most common commands.
- Specifies a fallback 'unknown_command' JSON object to be returned when the transcript does not match any known command — preventing random API calls from misrecognised speech.

12.6.3 Privacy Architecture

The A.U.R.A. AI stack is designed for zero data exfiltration:

- No audio is ever transmitted outside the local network. The INMP441 microphone feeds the ESP32; the ESP32 transmits only post-wake audio to the NAS on the same local subnet.
- whisper.cpp and Ollama run entirely on-device (NAS CPU/RAM). No OpenAI, Google, or Amazon API calls are made.
- Home Assistant runs locally; no cloud account is required (though optional Nabu Casa cloud integration is available if the user chooses).
- All service APIs are bound to the local network interface only — not exposed to the internet.

12.7 Home Assistant Integration

Integration	Trigger	Action	Protocol
A.U.R.A. source state → HA	ESP32 publishes source state on change (WIFI_ACTIVE, MULTIMEDIA_ACTIVE, IDLE)	HA automation triggers room scene based on source (e.g. PS2 active → dim lights to gaming mode)	MQTT publish from ESP32 → HA MQTT broker
HA → A.U.R.A. volume	HA automation or voice command via HA	HA calls ESP32 REST endpoint to set volume (requires digital potentiometer upgrade in v2.1)	HTTP REST to ESP32 web server
Room presence → HA	INMP441 presence detection event from ESP32	HA adjusts HVAC, lighting based on occupancy	MQTT publish
HA → audio notification	HA automation event (doorbell, alarm, timer)	HA sends TTS audio clip to Navidrome/Squeezelite for playback on A.U.R.A. speakers	HTTP POST to Squeezelite-ESP32 playlist endpoint
A.U.R.A. playback → HA media card	Navidrome / Squeezelite reports now-playing metadata	HA dashboard displays current track, album art, controls	MQTT or REST polling

12.8 Multi-Room Audio Expansion (Future)

The NAS-centric architecture is inherently multi-room capable. Adding a second A.U.R.A. node (ESP32 + PCM5102A + TPA3116D2 + speakers) in another room requires:

- The second ESP32 connects to the same WiFi network and registers with Lyrion Music Server (LMS) as a second Squeezelite player.
- LMS handles synchronised playback across both players using its built-in clock synchronisation — network jitter is absorbed by LMS's synchronisation buffer, keeping both rooms phase-aligned to within ~10 ms.
- CamillaDSP on the NAS can serve separate room correction profiles per player — each room's FIR filter is loaded independently based on the active Squeezelite player ID.
- Home Assistant manages room-independent volume and source selection via MQTT.

12.9 Bill of Materials — NAS Integration Subsystem

Qty	Component	Specification	Purpose	Status	Supplier
1	NAS server	x86-64 mini PC or purpose-built NAS — min 8 GB RAM, 1 GbE (see 12.2.1)	Hosts all software services	Planned v2.0	Local / online
1	NAS OS	Debian 12 LTS or TrueNAS Scale	Base OS for all Docker/bare-metal services	Planned v2.0	Free
2+	HDD (NAS storage)	2–4 TB 3.5" NAS-rated (WD Red / Seagate IronWolf)	RAID 1 media storage	Planned v2.0	Amazon / local
1	SSD (OS/apps)	128–256 GB SATA or NVMe	OS boot drive; Docker volumes; CamillaDSP filter files	Planned v2.0	Amazon
—	Jellyfin	Open source	Video + music media server	Planned v2.0	jellyfin.org
—	Navidrome	Open source	Lightweight music server (Subsonic API)	Planned v2.0	navidrome.org

Qty	Component	Specification	Purpose	Status	Supplier
—	CamillaDSP	Open source v2.0+	FIR room correction DSP engine	Planned v2.1	github.com/HEnquist/camilladsp
—	Lyrion Music Server	Open source (formerly Logitech Media Server)	Squeezelite multi-room backend	Planned v2.0	lyrion.org
—	whisper.cpp	Open source (MIT)	On-device speech recognition	Planned v2.1	github.com/ggerganov/whisper.cpp
—	Ollama + Llama 3 8B	Open source (Llama 3: Meta license)	Local LLM for voice command interpretation	Planned v2.1	ollama.com
—	Piper TTS	Open source (MIT)	On-device text-to-speech response	Planned v2.1	github.com/rhasspy/piper
—	Home Assistant	Open source (Apache 2.0)	Smart-room automation and device integration	Planned v2.1	home-assistant.io

12.10 Engineering Assessment — NAS Integration Subsystem

Strengths

- All services are open source and self-hosted — no recurring subscription costs, no cloud dependency, no vendor lock-in.
- CamillaDSP FIR room correction on NAS x86 hardware can run filter lengths (65,536 taps) that would be computationally impossible on the ESP32 or even a Raspberry Pi at real-time audio rates.
- Squeezelite + LMS multi-room synchronisation is a proven, mature protocol stack used in commercial audio products — reliable to < 10 ms inter-room alignment.
- Local AI stack (whisper.cpp + Ollama + Piper) is fully private — no audio leaves the local network at any point.
- Home Assistant integration provides a future-proof automation layer compatible with Zigbee, Z-Wave, WiFi, and hundreds of smart device integrations beyond audio.

Limitations / Risks

- NAS hardware cost is the largest single expenditure in the v2.x roadmap — a capable mini PC (Intel N305 or equivalent) with adequate RAM and storage is ₹15,000–₹40,000 depending on specification.
- CamillaDSP minimum-phase FIR latency of 20–40 ms means A.U.R.A. is not suitable as a low-latency monitor system. The Optical/Legacy (PS2/TV) path should bypass NAS DSP to avoid lip-sync issues.

- Local LLM inference (Ollama Llama 3 8B) on a low-power NAS CPU without GPU acceleration takes 3–8 seconds per response — acceptable for casual voice commands, but slower than cloud alternatives.
- whisper.cpp accuracy degrades in noisy environments and with non-native accents. ESP-SR wake-word detection is optimised for English — custom wake-word training is recommended for non-English users.
- Home Assistant introduces its own maintenance burden — updates, integrations, and automations require ongoing configuration. Not suitable for users unwilling to maintain the system.

Appendix A — Consolidated Bill of Materials

All subsystems merged into a single procurement table. Shared components (Cubetek, Optical DAC) listed once. v1.1 additions highlighted.

Layer	Component	Qty	Key Spec	Purpose	v1.1 Change	Supplier
WIFI	ESP32-WROOM-32 DevKit	1	240 MHz dual-core, WiFi+BT	Main MCU, WiFi streaming, source routing	—	Robu.in
WIFI	PCM5102A I2S DAC module	1	32-bit / 384 kHz / 112 dB SNR	WiFi path DAC — I2S to analog	—	AliExpress / Robu.in
WIFI	Ceramic cap 100 nF	2	50V	PCM5102A and ESP32 decoupling	—	Amazon
WIFI	10 kΩ series resistors	2	0.25W metal film	WiFi path output impedance isolation	NEW v1.1 — Supp E	Amazon
WIFI	RCA cable (shielded OFC)	1	0.5m	PCM5102A out to DPDT	—	Amazon
BT/OPT	Cubetek BT/Optical hub	1	BT 5.0 A2DP, Toslink out	BT receive + PS2 optical passthrough	—	Amazon / local
BT/OPT	Optical DAC (shared)	1	Toslink in, 192 kHz/24-bit, RCA out	Cubetek → analog for both BT and PS2	—	Amazon / AliExpress
BT/OPT	Toslink cable ×2	2	1m dust-capped	PS2→Cubetek, Cubetek→Optical DAC	—	Amazon
BT/OPT	10 kΩ series resistors	2	0.25W metal film	Multimedia path output impedance isolation	NEW v1.1 — Supp E	Amazon
BT/OPT	RCA cable (shielded OFC)	1	0.5m	Optical DAC out to DPDT	—	Amazon
ANALOG	TPA3116D2 amp board	1	19V, 50W+50W, heatsink, volume pot	Class-D speaker amplifier	—	Amazon / Robu.in
ANALOG	DPDT toggle switch	1	> 1A rated	L+R simultaneous source selection	—	Amazon / local
ANALOG	DC power jack female	1	5.5mm × 2.1mm	19V adapter input	—	Amazon
ANALOG	DC splitter	1	5.5mm × 2.1mm 1-to-2	Split 19V to amp and buck	—	Amazon

Layer	Component	Qty	Key Spec	Purpose	v1.1 Change	Supplier
ANALOG	Speaker wire	2m	18 AWG OFC 2-conductor	TPA3116D2 to Sony MHC-GX2000	—	Local
PDN	LM2596 buck module	1	Adjustable, multiturn pot, 3A	19V → 5V brain rail	—	Amazon
PDN	19V adapter	1	Minimum 4A / 76W	Main power source	4A minimum — revised from 3A. Supp C	Local / online
PDN	Electrolytic cap	1	1000 μ F 25V low-ESR	19V muscle rail bulk reservoir	—	Robu.in
PDN	Electrolytic cap	2	100 μ F 16V	LM2596 output filter	—	Robu.in
PDN	Ceramic cap 100 nF	10+	50V	MCU/DAC local decoupling	—	Amazon
PDN	Ferrite bead (optional)	2	10 μ H	PCM5102A VCC noise filter	—	Robu.in
PDN	Wire (muscle rail)	1m	18 AWG silicone	High-current power paths	—	Local
PDN	Wire (signal GND)	1m	24 AWG	Star ground network	—	Local
ROUTING	Resistors (divider)	4	10 k Ω + 3.3 k Ω ($\times 2$ channels)	Signal-presence ADC voltage dividers	—	Amazon
ROUTING	LED 5mm blue + green	2	5mm	Source active indicators	—	Amazon
ROUTING	Resistor 220 Ω	2	0.25W	LED current limiting	—	Amazon
ROUTING v2	SPDT relay module	1	5V coil, > 1A contacts	Automatic source switching (v2.0)	—	Amazon / Robu.in
ROUTING v2	2N2222 transistor	1	NPN	Relay coil driver (v2.0)	—	Amazon
ROUTING v2	1N4007 diode	1	1A	Relay flyback protection (v2.0)	—	Amazon

Appendix B — Pin Mapping and Wiring Tables

B.1 ESP32 Complete Pin Assignment

GPIO	Function	Connected To	Direction	Notes
GPIO 22	I2S DOUT	PCM5102A DIN (Pin 14)	Output	Serial audio data, MSB first
GPIO 25	I2S LRCLK	PCM5102A LRCK (Pin 15)	Output	Word clock — L/R channel select
GPIO 26	I2S BCLK	PCM5102A BCK (Pin 13)	Output	Bit clock for I2S data
GPIO 27	PCM5102A XSMT	PCM5102A XSMT (Pin 17)	Output	HIGH = play, LOW = soft mute
GPIO 34	WiFi ADC monitor	10k+3.3k divider from PCM5102A RCA	Input only	ADC1_CH6 — no internal pullup. Threshold derivation: Supplement A
GPIO 35	Multimedia ADC monitor	10k+3.3k divider from Optical DAC RCA	Input only	ADC1_CH7 — no internal pullup. Threshold derivation: Supplement A
GPIO 2	WiFi active LED	220 Ω + blue LED to GND	Output	HIGH = WiFi source active
GPIO 4	Multimedia active LED	220 Ω + green LED to GND	Output	HIGH = Multimedia source active
GPIO 5	Relay control (v2.0)	2N2222 base via 1 k Ω → relay coil	Output	HIGH = Optical DAC position selected
3.3V	PCM5102A logic VCC	PCM5102A VCC (Pin 1) via ferrite bead	Power out	ESP32 onboard 3.3V regulator
VIN / 5V	ESP32 power in	LM2596 5V brain rail output	Power in	5V from buck converter
GND	Common ground	Star ground node (LM2596 Vout GND)	Ground	Must be single-point star

B.2 PCM5102A Module Pin Assignment

PCM5102A Pin	Signal	Connected To	Notes
Pin 1 — VCC	3.3V supply	ESP32 3.3V pin (via 10 μ H ferrite + 100 nF)	Star GND return via Pin 2. Ferrite bead added for LM2596 noise isolation.
Pin 2 — GND	Ground	Star GND node	Via short direct wire
Pin 13 — BCK	Bit clock	ESP32 GPIO 26	I2S bit clock
Pin 14 — DIN	Data in	ESP32 GPIO 22	I2S audio data

PCM5102A Pin	Signal	Connected To	Notes
Pin 15 — LRCK	Word clock	ESP32 GPIO 25	I2S L/R channel select
Pin 12 — SCK	System clock	3.3V rail (tied HIGH)	Enables internal PLL — no external MCLK needed
Pin 16 — FMT	Format select	GND (tied LOW)	LOW = I2S standard format
Pin 8 — DEMP	De-emphasis	GND (tied LOW)	LOW = disabled
Pin 17 — XSMT	Soft mute	ESP32 GPIO 27	HIGH = play; float or HIGH on boot
RCA Out	Analog L/R output	10 kΩ series R → DPDT switch Position A (NEW v1.1)	Series R added for output impedance isolation — Supplement E

B.3 Power Rail Summary

Rail	Voltage	Source	Loads	Wire AWG
Muscle rail	19V DC	Laptop adapter direct (min 4A — Supplement C)	TPA3116D2 amplifier only	18 AWG (short run)
Brain rail	5V DC regulated	LM2596 output	ESP32, PCM5102A, Optical DAC, Cubetek	22 AWG
Logic rail	3.3V DC	ESP32 onboard regulator	PCM5102A logic supply (via ferrite bead)	24 AWG (very short)
Star GND	0V reference	DC barrel jack GND	All components — single star point	18 AWG (muscle) / 24 AWG (signal)

B.4 Series Resistor Placement (NEW v1.1)

Resistor	Value	Placed In Series On	Between	And	Purpose
R1 (WiFi L)	10 kΩ	PCM5102A Left RCA hot wire	PCM5102A RCA L output	DPDT Switch Position A Left input lug	Output impedance isolation — Supplement E
R2 (WiFi R)	10 kΩ	PCM5102A Right RCA hot wire	PCM5102A RCA R output	DPDT Switch Position A Right input lug	Output impedance isolation — Supplement E
R3 (Multimedia L)	10 kΩ	Optical DAC Left RCA hot wire	Optical DAC RCA L output	DPDT Switch Position B Left input lug	Output impedance

Resistor	Value	Placed In Series On	Between	And	Purpose
					isolation — Supplement E
R4 (Multimedia R)	10 kΩ	Optical DAC Right RCA hot wire	Optical DAC RCA R output	DPDT Switch Position B Right input lug	Output impedance isolation — Supplement E